

SAS演習

NCD Epidemiology Research Center (NERC)
Department of Medical statistics

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0. Course series orientation

コースの進め方

- 本コースの目的

- 医学研究における統計的手法を理解し、活用すること
- 臨床研究よりも疫学研究での応用に重点を置いて講義する。

- 各講義(パート)

- シリーズ前半(基礎編 Fundamental)

- 統計パッケージを用いて、基本的な統計手法の復習を行う。

- シリーズ後半(*応用編 Advances)

- 疫学研究論文で使用されている統計手法の例を通して、研究テーマやデータの特徴に適した統計手法を学びます。

*受講はオプション

コース構成

本講座は、以下の11回に分けて実施します。

I Fundamental シリーズ

- 1 イントロ
- 2 2群の比較 (Continuous and Categorical data)
- 3 一般線形モデル(1) 回帰分析
- 4 一般化線形モデル ロジスティック回帰, Poisson回帰
- 5 生存時間析
- 6 主成分分析, 因子分析, クラスタ分析, 潜在クラス分析

II Advanced シリーズ

- 7 一般線形モデル (2) ANOVA 混合モデル
- 8 Reporting Guide: PRISMA Meta-analysis
- 9 Reporting Guide: CONSORT/STROBE
- 10 欠測データの解析, 相関のあるデータの解析: 一般化線形混合モデル
- 11 因果推論入門 (非巡回有向グラフ(DAG), 傾向スコア, mediation analysis)
～ SAS Causaltrt/Causalsmed/Causalgraph/PSmatch ～

Recommendations for Statistical Reporting in Cardiovascular Medicine. Circulation. 2021;144:e70–e91

Circulation

AHA SPECIAL REPORT

Recommendations for Statistical Reporting in Cardiovascular Medicine

A Special Report From the American Heart Association

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ABSTRACT: Statistical analyses are a crucial component of the biomedical research process and are necessary to draw inferences from biomedical research data. The application of sound statistical methodology is a prerequisite for publication in the American Heart Association (AHA) journal portfolio. The objective of this document is to summarize key aspects of statistical reporting that might be most relevant to the authors, reviewers, and readership of AHA journals. The AHA Scientific Publication Committee convened a task force to inventory existing statistical standards for publication in biomedical journals and to identify approaches suitable for the AHA journal portfolio. The experts on the task force were selected by the AHA Scientific Publication Committee, who identified 12 key topics that serve as the section headers for this document. For each topic, the members of the writing group identified relevant references and evaluated them as a resource to make the standards summarized herein. Each section was independently reviewed by an expert reviewer who was not part of the task force. Expert reviewers were also permitted to comment on other sections if they chose. Differences of opinion were adjudicated by consensus. The standards presented in this report are intended to serve as a guide for high-quality reporting of statistical analyses methods and results.

Key Words: biostatistics

The American Heart Association (AHA) journal portfolio includes *Arteriosclerosis, Thrombosis, and Vascular Biology*; *Circulation*; *Circulation: Arrhythmia and Electrophysiology*; *Circulation: Genomic and Precision Medicine*; *Circulation: Cardiovascular Imaging*; *Circulation: Cardiovascular Interventions*; *Circulation: Cardiovascular Quality and Outcomes*; *Circulation: Heart Failure*; *Circulation Research*; *Hypertension*; *Journal of the American Heart Association*; and *Stroke*. Collectively, these journals receive >20 000 manuscript submissions per year. The statistical analysis methods used in these submitted manuscripts vary in terms of approaches and sophistication, ranging from simple descriptive statistics (eg, counts and frequencies) to complex modeling (eg,

multivariable regression, propensity score adjustment, instrumental variables, survival analyses, development of risk prediction models) and meta-analyses. Most original research manuscripts submitted to the AHA journal portfolio are reviewed by at least 1 statistical reviewer or a statistical editor before publication. The AHA Scientific Publication Committee oversees the AHA journal portfolio.

TASK FORCE CHARGE AND APPROACH

To harmonize expectations for reporting statistical analyses across journals, the AHA Scientific Publication Committee charged the task force with the following:

given manuscript. The 12 sections are as follows:

1. General Standards
2. Observational Studies: Diagnostic Tests and Validation
3. Observational Studies: Clinical Prediction Models
4. Statistical Genetics
5. Randomized Controlled Trials
6. Systematic Reviews and Meta-Analyses
7. Survival Analyses
8. Bayesian Statistical Approaches
9. Missing Data
10. Correlated Data
11. Covariable Adjustment and Propensity Scores
12. Power and Sample Size Considerations

1.1. Existing Guidelines

The AHA journals encourage authors to follow relevant existing guidelines when applicable (<https://www.aha->

Table. Reporting Guidelines by Study Type

Study type	Reporting guideline	Additional Info
Study protocols	SPIRIT	PRISMA-P
Observational studies	STROBE	Extensions
Diagnostic/prognostic studies	STARD	TRIPOD
Randomized trials	CONSORT	Extensions
Systematic reviews	PRISMA	Extensions

For more information, see the EQUATOR network.¹ CONSORT indicates Consolidated Standards of Reporting Trials; EQUATOR, Enhancing the Quality and Transparency of Health Research; PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses; PRISMA-P, Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols; SPIRIT, Standard Protocol Items: Recommendations for Interventional Trials; STARD, Standards for Reporting of Diagnostic Accuracy Studies; STROBE, Strengthening the Reporting of Observational Studies in Epidemiology; and TRIPOD, Transparent Reporting of a Multivariable Prediction Model for Individual Prognosis or Diagnosis.

Advanceシリーズでは、Circulation CVD statガイドラインの項目を参考に、FundamentalシリーズやReportingガイドラインに記載されていない項目を取り上げて、項目の充実に努めます。

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Writing Group and Reviewer Disclosures are available with this article at <https://www.ahajournals.org/doi/suppl/10.1161/CIRCULATIONAHA.121.055393>.
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本講座の受講方法、単位について

◆講義形式: オンデマンド

- 前半は講義資料をもとに学習し、後半は講義資料をもとに各自で演習に取り組みます。

◆Login address: Posted on Web Class

Webクラスは講義テーマごとに分け、教材や動画のログインアドレスも提供しますので、ご自分のペースで受講してください。

◆Earn credits:

各講義ごとにレポートがあります

取り組み、Webclassから提出するようにしてください

◆SAS, R, SPSS, STATA などの統計解析ソフトを使用してもかまいません。いずれかを各自で用意してください。

授業に関して何か質問があれば、いつでも連絡してください。コースについて何か質問がありましたら、いつでもメールまたはWebclassからお気軽にご連絡ください。

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Webclass

This lecture will be given using SAS.

- R, SPSS user

- SPSS, Rを使用する場合は、別途補足資料を参照の上、課題を行ってください(SPSSは準備中)。

- Not full version of SAS user

- SAS製品版を利用できないが、SASを使ってみたいという方には、SAS Academic Editionの利用をお勧めします。
- SAS Academic Editionはインターネットに接続できる環境であれば利用できますので、ユーザー登録をしてご利用ください。
- また、ウェブクラスで入門用のサプリメントを作成しています。
- SAS Academic Editionは製品版と比較して機能が制限されていますが、本講義で使用する解析はこれでカバーできると思います。

JMPやSTATAなど他のソフトを使う場合は、相談してください。

本講座のオリエンテーションはこれで終了
です。

講義や演習に関するご質問は
いつでもどうぞ。



1 Introduction

Using the Statistical package

Goals of Todays Lectures:

統計パッケージの使用に慣れる

- 1) データをパッケージにアップロードできる
- 2) データの基本特性を様々なプロシジャーで把握できるようになる

Today's overview

1. 統計パッケージの使用法

- 1) 統計パッケージSASのイントロダクション
- 2) データの読み込み

2. 記述統計

- 1) 位置: 平均, メディアン, パーセント
- 2) ばらつき: range, inter-quartile range, CV
- 3) 分布: 歪度skewness, 尖度kurtosis, 正規性
- 4) 2変量の相関: Correlation

グラフ

Software packages for statistical analysis of data

- SAS
- R
- Python
- STATA

Programming statement (code) type

- SPSS
- JMP

Point-and-click interface

複数のパッケージが使用できることが望ましい ！

NCD Reader Course students are encouraged to use SAS

Statistical Analysis System (SAS)

-Strength-

- 広く用いられている (in Business)
- 製薬企業にとっては申請データの解析・作成で用いられている (New Drug Application to the FDA)
- プログラムを共有財産として活用可能
- プログラムの妥当性が検証されている。
R: program supply are by user-based (妥当性の面ではSASに軍配)
- データハンドリングに強い (あらゆる形式のデータに対しても柔軟に対応)
- 豊富な統計手法が用意されている行列言語が利用可能
- AIのプラットフォームもあり
- ユーザーのコミュニティ活動活発

-Weakness-

- プログラミングコードを使用するため、初期ユーザーへのパッケージの導入・教育に時間がかかる。
- ライセンスコストが高い
- 大学在学中は利用できるが、卒業後は利用できなくなることが多い。

R, Python等のメリット

メリット

- Free
 - どの環境でも使用できる
 - コード配布などの面でメリット
 - 書籍、論文等での配布例多い
- ユーザーが多い、書籍等豊富
- AI系の解析への拡張性が高い
- 新しい解析への対応が早い
- Rについては、RstudioやEZR(point-click型)など使用しやすい統合インターフェイス環境が整ってきている

デメリット

- 新しいプログラムのValidation
 - 近年は公開にあたりチェック機能が充実

自分のPCでパッケージを使用するには SAS

- SAS OnDemand Academics (無償)
 - https://www.sas.com/ja_jp/software/on-demand-for-academics.html
 - 登録が必要
 - クラウド環境



サインイン

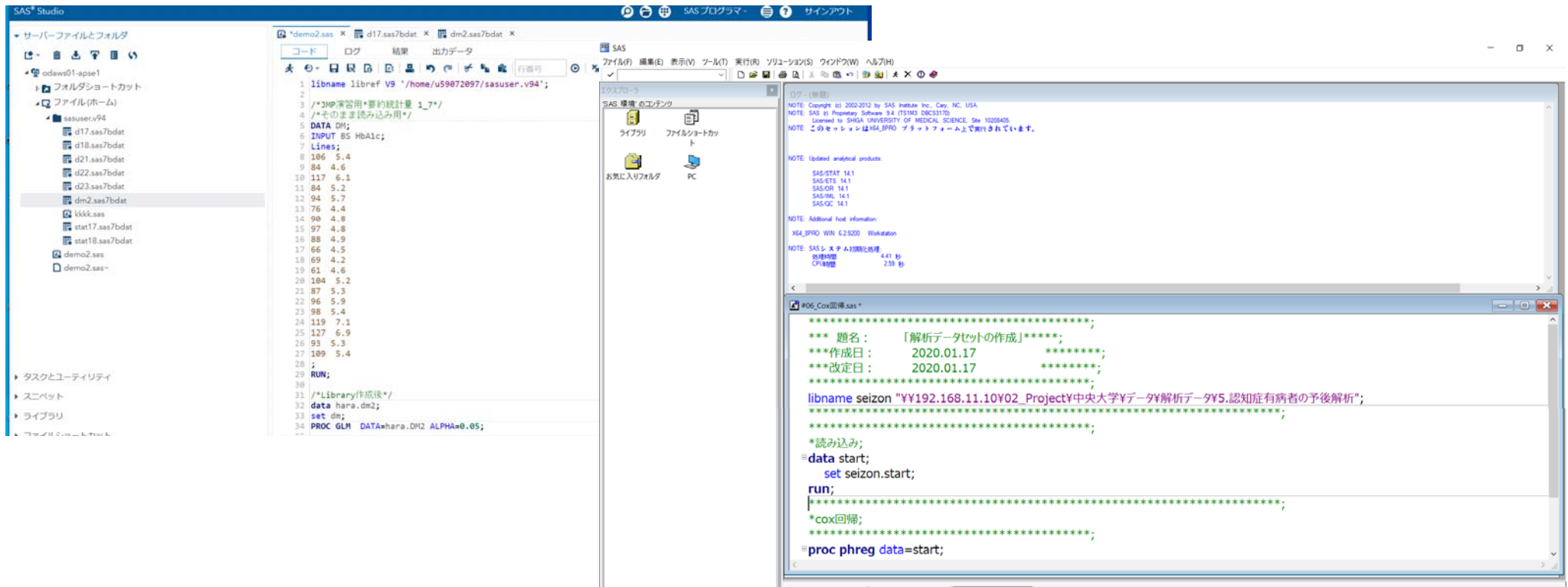
- ① SAS プロファイルを登録
- ② SAS OnDemand の登録。
- ③ SAS Studio の利用。

SAS studio画面

- ①コード：プログラムを各画面
- ②ログ、③結果、④データセット画面

On-demand版【コード】画面

SASパッケージ版(有料)の画面



1) A brief Introduction to SAS

統計ソフト（パッケージ）の使用法

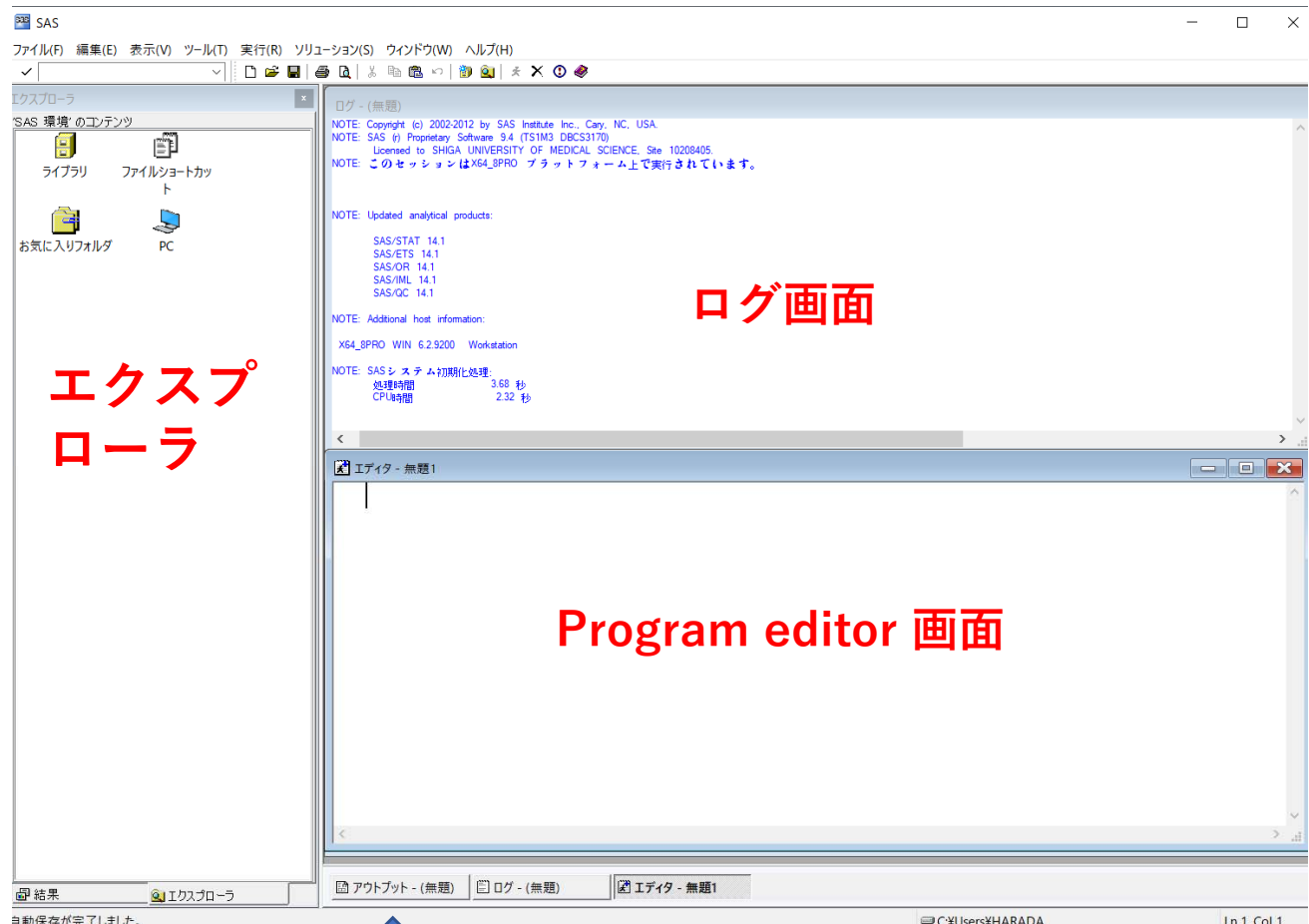
SASを起動してみよう



Double-click the SAS icon, and SAS will launch.

Display Manager System(DMS)は、多くのウィンドウを持ち、連動してSASシステムを操作することができる。

Fig. SAS Startup Screen



エクス
プローラ

ログ画面

Program editor 画面

Output window

Basic operation of SAS environment

主なウィンドウの使い方は次のとおり

1. プログラム・エディタ・ウィンドウ

SASプログラムの編集

2. ログウィンドウ

SASプログラムの実行経過に関するメッセージなどの情報を表示する

3. 出力ウィンドウ

SASプログラムの実行結果を表示する

4. Resultウィンドウ

SASプログラムの実行状況を確認することができる

5. エクスプローラウィンドウ

ライブラリ等で実行中のSASデータセットを確認することができます。

ウィンドウ自体の大きさや位置は、状況に応じて変更可能

各ウィンドウは、マウスで領域をクリックすることで起動し、各領域で作業できる

SAS Data set

- SASのデータセットは、変数×観測値 の長方形の形をしており、多変数データの基本的なパターンである

SAS Data Set Example 1

ID	年齢	性	治療群	改善度	除外・脱落コード
1	48	F	1	1	0
2	52	M	2	1	.
3	64	F	1	2	0
4	60	M	1	1	2

変数の種類は2つ。
数値型と文字型

”.[ピリオド]”は欠損値を示す。

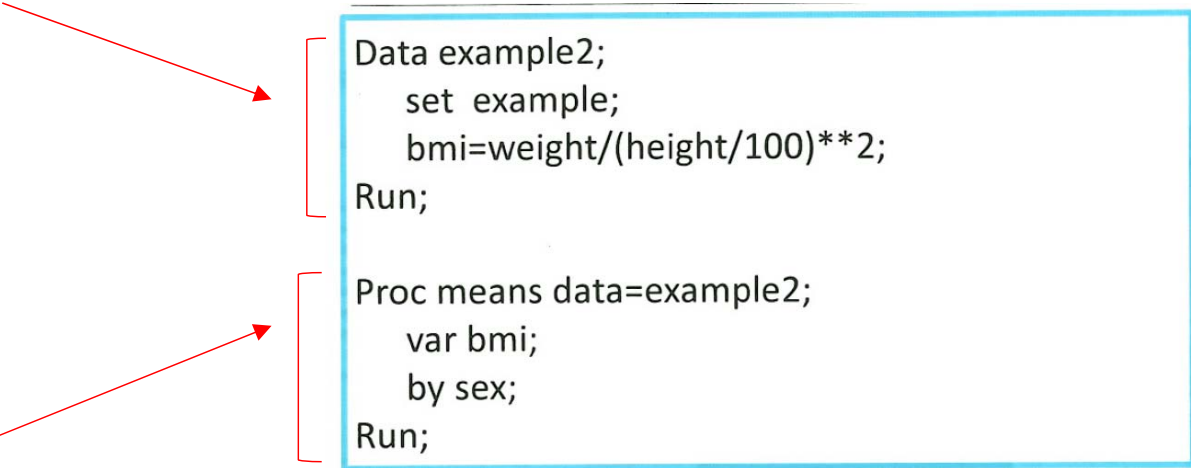
SAS Data Set Example 2

ID	測定日	薬量	最大血圧	最小血圧
1	90/02/04	10	165	109
1	90/02/05	20	155	97
1	90/02/06	20	154	90
2	90/01/09	20	138	101
2	90/01/10	20	129	89

SAS program structure

□ Data step

- Statement “**Data**.....; **Run**;
- Data loading, direct input, processing, generation, etc.



```
Data example2;  
  set example;  
  bmi=weight/(height/100)**2;  
Run;
```

```
Proc means data=example2;  
  var bmi;  
  by sex;  
Run;
```

□ PROC step

- Statement “**PROC**.....; **Run**;
- Call each procedures and perform statistical analysis.

2) Reading data

SASにおけるデータ読みこみ方法は様々ある

Reading Data ① Direct

- 直接入力する:

- INPUT文で、変数名を列挙
- 文字列データは、変数名の後\$をつける.

*SAS のデータセットと変数名は、最初の文字が数字であってはいけない

- CARDS(またはdataline)ステートメントの後に、生データを記載する

DATA Step Example

```
data test;
```

```
input id age sex $ arm effect drop;
```

```
cards;
```

1	48	F	1	1	0
2	52	M	2	1	0
3	64	F	1	2	0
4	60	M	1	1	2

```
;
```

```
run;
```

All SAS programming statements end with a semicolon (;).

DATA Step Example

data test;

input id age sex \$ arm effect drop;

cards;

1 48 F 1 1 0

2 52 M 2 1 0

3 64 F 1 2 0

4 60 M 1 1 2

;

run;

Reading Data ② import external files

- 外部ファイルを読み込んでSASデータセットに加工するには、次のDATAステップをコーディングします。
- INFILE文の中で一重引用符で囲んだデータファイルをインポート。
- ファイルの場所には注意してください。

```
data lab;  
  infile 'C:\lab.csv' delimiter=',' firstobs=2;  
  input age weight height h_bp l_bp tp got gpt alp ggpt bun;  
  age10=int(age/10);  
run;
```

「lab.csv」

• Cドライブにあるlab.csvを読み込む。
csv形式は「delimiter=','」を指定してください。
データ開始行を "firstobs" で指定する。

	A	B	C	D	E	F	G	H	I	J	K
1	age	weight	height	h bp	l bp	tp	got	gpt	alp	ggpt	bun
2	43	61	147.5	110	78	7.6	19	14	4.4	10	11
3	82	56	144.3	158	70	7.6	27	20	5.9	7	15
4	40	46.5	146	110	76	7.3	14	12	3	8	16
5	71	57.5	146.1	186	118	8.5	22	14	7.5	22	15
6	50	58	156.5	160	98	7.3	15	13	5.9	8	14
7	53	53.5	144.8	150	90	7.6	25	20	6.9	12	13
8	54	50	139.3	164	100	7.1	20	17	4.2	23	18
9	76	40	136.5	130	70	7.7	24	16	4.7	12	19
10	62	48	151.9	114	76	6.8	20	17	5.4	10	14
11	62	49	154.6	142	88	8	24	17	5.5	16	13
12	48	48.5	150.2	108	70	7.4	14	14	4.7	9	12

options in the INFILE statement

- **Delimiter =(DLM =)**

Specify this when reading data whose variables are separated by characters other than single-byte spaces.

.csv comma DLM = ‘,’

.txt tab DLM = ‘09’x

 space DLM = ‘ ’

Reading Data ③ SAS data set files

SASデータセットには、次の2種類がある

- 1) 保存された(永続的な)SASデータセット
ライブラリに保存されたデータセット
SASを終了しても消えない
- 2) 一時的なSASデータセット
ワークライブラリのSASデータセット
SASを終了すると消滅

Reading Data ③ SAS data set files

- SAS LIBRARYの指定
 - LIBENAME statement
 - LIBENAME *library name* 'save folder location'
 - Libname mysas 'c:\¥*****¥*****¥*****';
- Loading a stored SAS data set
 - SET statement
 - SET library name. SAS dataset;

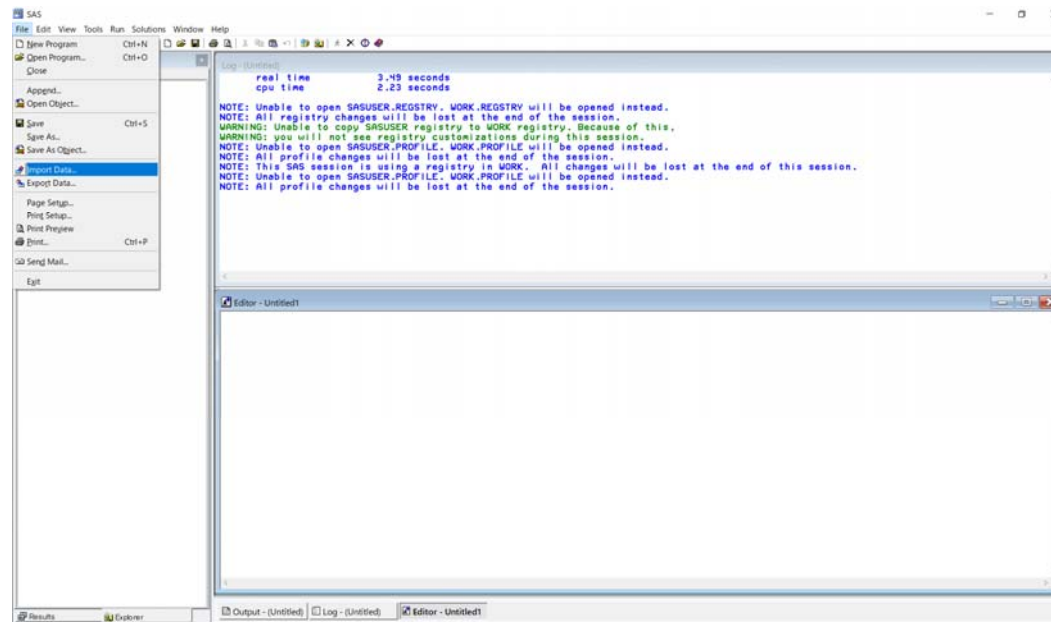
```
Data data2;  
      set mysas.data1;  
run;
```

左の例
mysasライブラリに格納されているデータセット "data1" をロードし、"data2" に設定する。

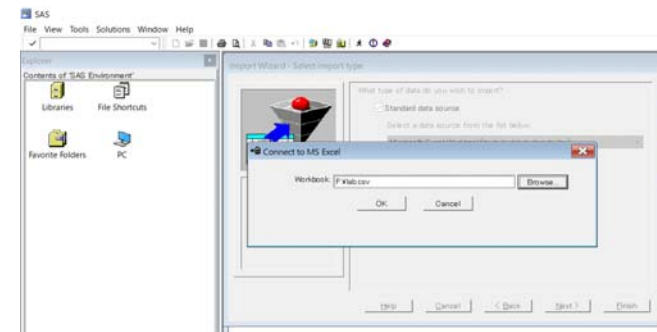
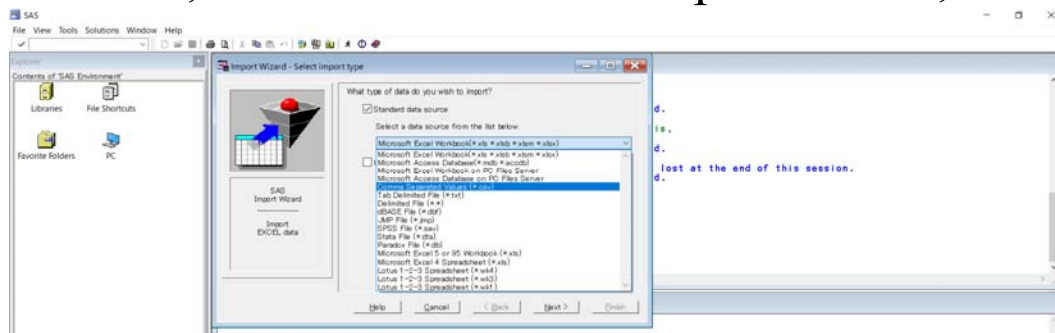
Reading Data ④ IMPORT wizard

- The following Excel data (file name lab.csv).

1. FILE→Import



2. Next, select "CSV" in the import wizard, and select the file location and sheet



The screenshot displays the SAS software interface. In the foreground, the 'Import Wizard - Select library and member' dialog box is open. It features a checkered flag icon on the left and a 'Choose the SAS destination:' section on the right. The 'Library' dropdown is set to 'WORK' and the 'Member' dropdown is set to 'lab2'. At the bottom of the dialog are buttons for 'Help', 'Cancel', '< Back', 'Next >', and 'Finish'.

In the background, the SAS Explorer window is visible on the left, showing the 'Contents of SAS Environment' with icons for Libraries, File Shortcuts, Favorite Folders, and PC. On the right, a code editor shows the following SAS code:

```
d.
ls,
d.
lost at the end of this session.
d.
```

In Temporary Work library, "lab2" dataset was made

データの読み込みは
できそうですか？



PROC Step (SUBMIT)

- PROCステップとは、PROC文で始まり、RUN文で終わる一連のプログラムのことです。
- プロシージャごとに異なるオプションやステートメントがたくさんあり、それぞれのプロシージャを覚える必要があります(参照する必要があります)。

```
PROC ProcedureName Option;  
    Statements....;  
RUN;
```

Here is the **GLOBAL STATEMENT syntax** common to all procedures.

DATA = SAS data set name (if not specified, use the previous one)

VAR statement Specifies the variable to use

BY statement Perform processing by group

WHERE statement Limit the observations of interest



演習 #1

データセットを4つの方法で作成してみる。

プログラムは "dataset.txt "に示されており、これを参照し、データの読み方として4つの方法を練習してください。

Reading data methods

- 1) Direct
- 2) Infile
- 3) SAS dataset
- 4) Import wizard

Today's overview

1. 統計パッケージの使用法

- 1) 統計パッケージSASのイントロダクション
- 2) データの読み込み

2. 記述統計

- 1) 位置: 平均, メディアン, パーセント
- 2) ばらつき: range, inter-quartile range, CV
- 3) 分布: 歪度skewness, 尖度kurtosis, 正規性
- 4) 2変量の相関: Correlation

グラフ

The distributions of numeric variables

SASを使って、以下のような指標を評価してみます。

位置：平均値、中央値、パーセンタイル

分散：範囲、四分位範囲、CV

分布：歪度, 角化度, 正規性

2変数間の関連性 相関

例題

- Health survey data
 - N=421
 - Age 40-60yrs.
 - Variable
 - SEX, AGE, Systolic Blood Pressure (SBP), Diastolic blood pressure (DBP)
 - Urinary Sodium (SALT)
 - Dataset name=health

Dataset health (health_data.txt)

```
*****
```

```
01_health_data.txt
```

```
*****;
```

```
/*data description "health data" */
```

```
data health; /*dataset name*/
```

```
input sex $ age sbp dbp salt; /*"$" indicates character variable*/
```

```
cards;
```

```
F 64 137 86 13.4
```

```
F 55 97 66 8.9
```

```
F 52 151 93 12.7
```

```
F 63 105 75 10.4
```

```
F 66 139 71 6.9
```

```
F 48 132 85 10.1
```

```
F 53 129 67 7.4
```

```
F 62 138 85 8.0
```

```
.
```

```
.
```

```
;
```

```
/* example of retain statement*/
```

```
data health2;
```

```
  set health;
```

```
  retain id 0;
```

```
  id=id+1;
```

```
run;
```

データセット "health"にはユニークIDがない。
→データセット "health2"は、"health"に新たにID
変数を追加しています

Check the loaded dataset with "Proc print".



Q1. Create Dataset

Create a Dataset "health" to "health2".

"health"

The SAS System

Obs	sex	age	sbp	dbp	salt
1	F	64	137	86	13.4
2	F	55	97	66	8.9
3	F	52	151	93	12.7
4	F	63	105	75	10.4
5	F	66	139	71	6.9
6	F	48	132	85	10.1
7	F	53	129	67	7.4
8	F	62	138	85	8.0
9	F	64	126	75	10.2
10	F	47	101	73	9.3
11	F	67	133	65	9.2
12	F	50	97	60	7.9
13	F	58	104	69	10.6
14	F	61	109	65	9.4
15	F	55	117	69	10.1

SAS code

```
data health2;  
set health;  
retain id 0;  
id=id+1;  
run;
```

"health2"

The SAS System

Obs	sex	age	sbp	dbp	salt	id
1	F	64	137	86	13.4	1
2	F	55	97	66	8.9	2
3	F	52	151	93	12.7	3
4	F	63	105	75	10.4	4
5	F	66	139	71	6.9	5
6	F	48	132	85	10.1	6
7	F	53	129	67	7.4	7
8	F	62	138	85	8.0	8
9	F	64	126	75	10.2	9
10	F	47	101	73	9.3	10
11	F	67	133	65	9.2	11
12	F	50	97	60	7.9	12
13	F	58	104	69	10.6	13
14						
15						

Create a variable "id"
of identification

Use Univariate procedure

形状、位置、分布の指標を確認

1. Location

- 1) Mean
- 2) Median
- 3) Mode
- 4) Percentile

2. Variability dispersion

- 1) Variance
- 2) SD
- 3) Range
- 4) Inter-quartile range

3. Shape

- 1) Skewness 歪度
- 2) Kurtosis 尖度

PROC univariate ① code

- PROC UNIVARIATE

```
proc univariate option data=dataset name;  
  var variable list;  
  [by stratified variable;]  
run;
```

PROC univariate ②

```
proc univariate data=health2
  var age;
run;
```

Variability dispersion

- 1) Variance
- 2) SD
- 3) Range
- 4) Inter-quartile range

Shape

- 1) Skewness
- 2) Kurtosis

Location

- 1) Mean
- 2) Median
- 3) Mode
- 4) Percentile

The SAS System

The UNIVARIATE Procedure
Variable: age

Moments			
N	421	Sum Weights	421
Mean	58.9049881	Sum Observations	24799
Std Deviation	7.81576544	Variance	61.0861893
Skewness	-0.8350078	Kurtosis	-0.3284162
Uncorrected SS	1486441	Corrected SS	25656.1995
Coeff Variation	13.2684271	Std Error Mean	0.3809171

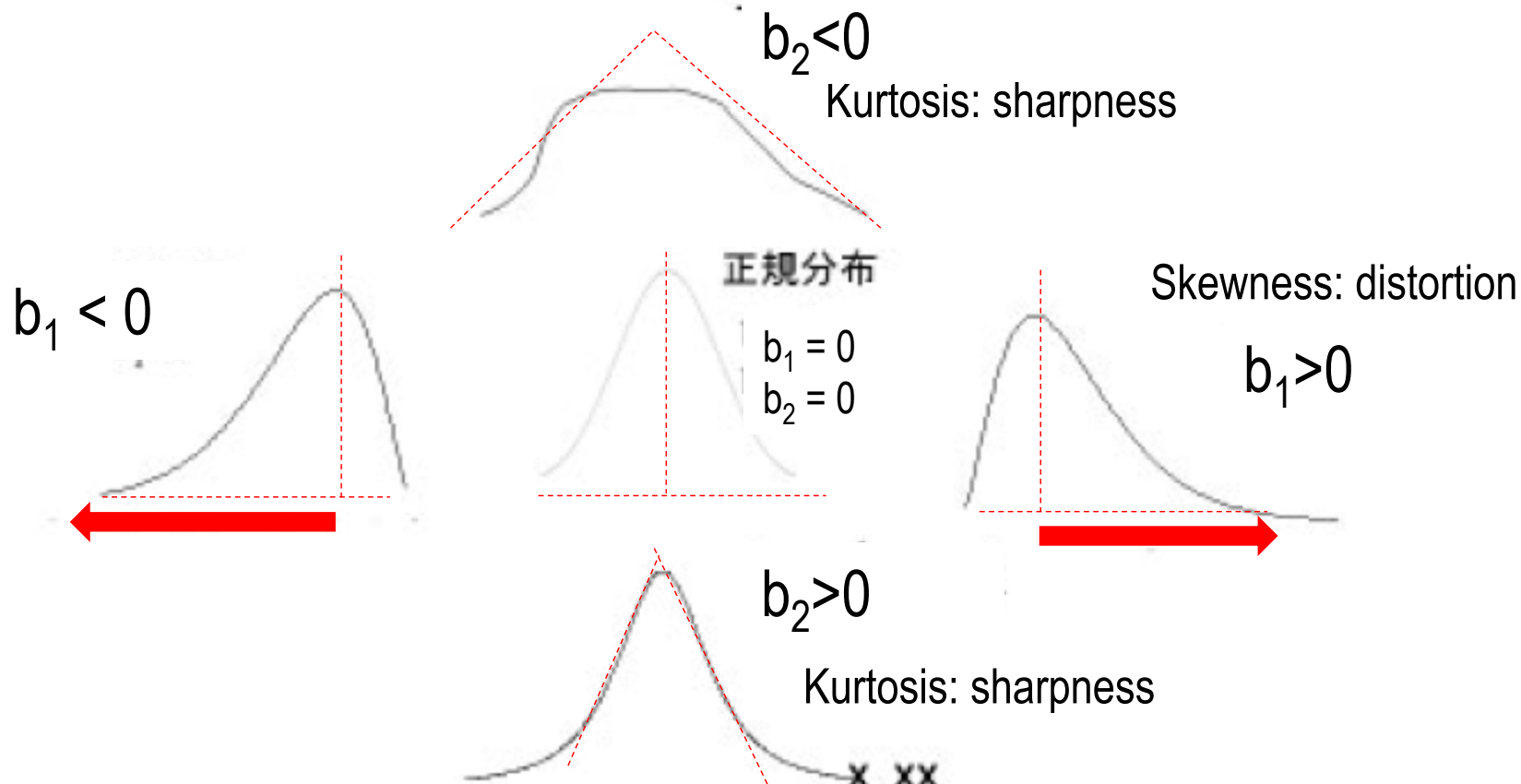
Basic Statistical Measures			
Location		Variability	
Mean	58.90499	Std Deviation	7.81577
Median	61.00000	Variance	61.08619
Mode	65.00000	Range	29.00000
		Interquartile Range	11.00000

Tests for Location: Mu0=0				
Test	Statistic		p Value	
Student's t	t	154.6399	Pr > t	<.0001
Sign	M	210.5	Pr >= M	<.0001
Signed Rank	S	44415.5	Pr >= S	<.0001

Quantiles (Definition 5)	
Level	Quantile
100% Max	69
99%	69
95%	68
90%	67
75% Q3	65
50% Median	61
25% Q1	54
10%	46
5%	42
1%	40
0% Min	40

PROC univariate ③Shape

歪度 Skewness(b_1), 尖度 Kurtosis(b_2)



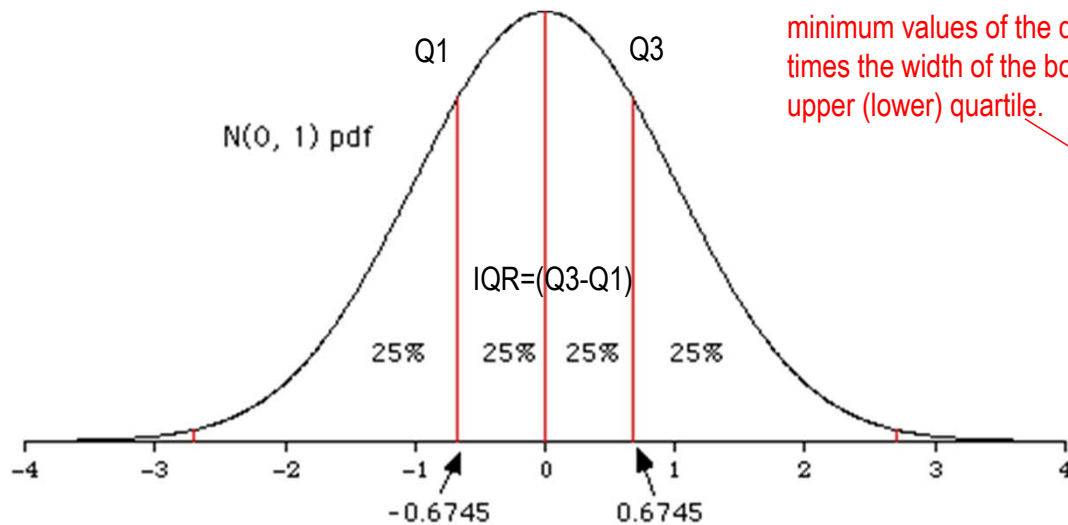
Skewness: distortion

Kurtosis: sharpness

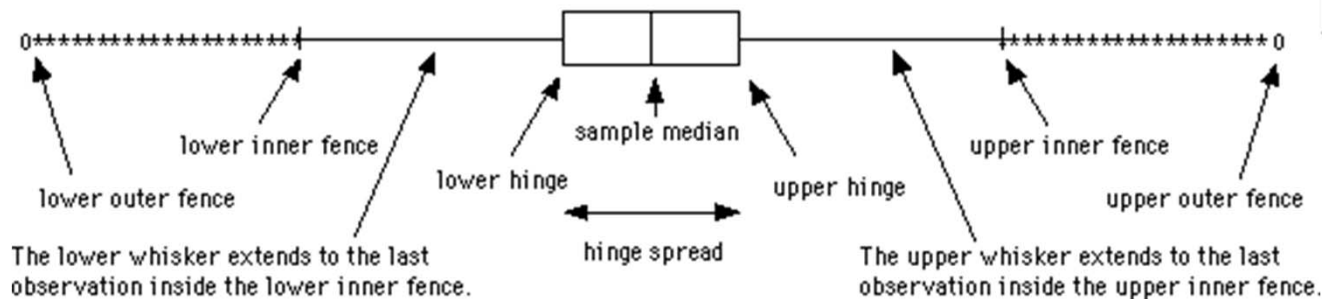
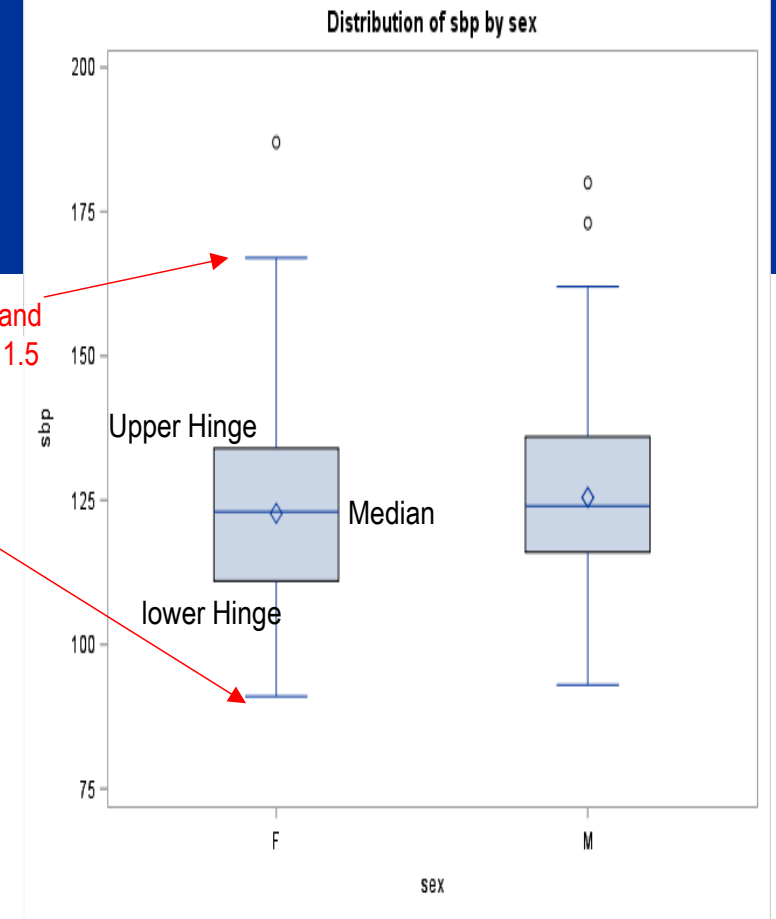
$$b_1 = \frac{\sum_{i=1}^n (X_i - \bar{X})^3 / (n-1)}{SD^3}, \quad b_2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^4 / (n-1)}{SD^4} - 3$$

PROC univariate

④ Boxplot 箱ひげ図



Extreme values are the maximum and minimum values of the data within 1.5 times the width of the box from the upper (lower) quartile.



```
proc boxplot
data=health;
plot sbp*sex/
boxstyle=schematic;
run;
```

■

Q2 Use Univariate procedure

形状、位置、分布指標を確認する (var= SBP, DBP)

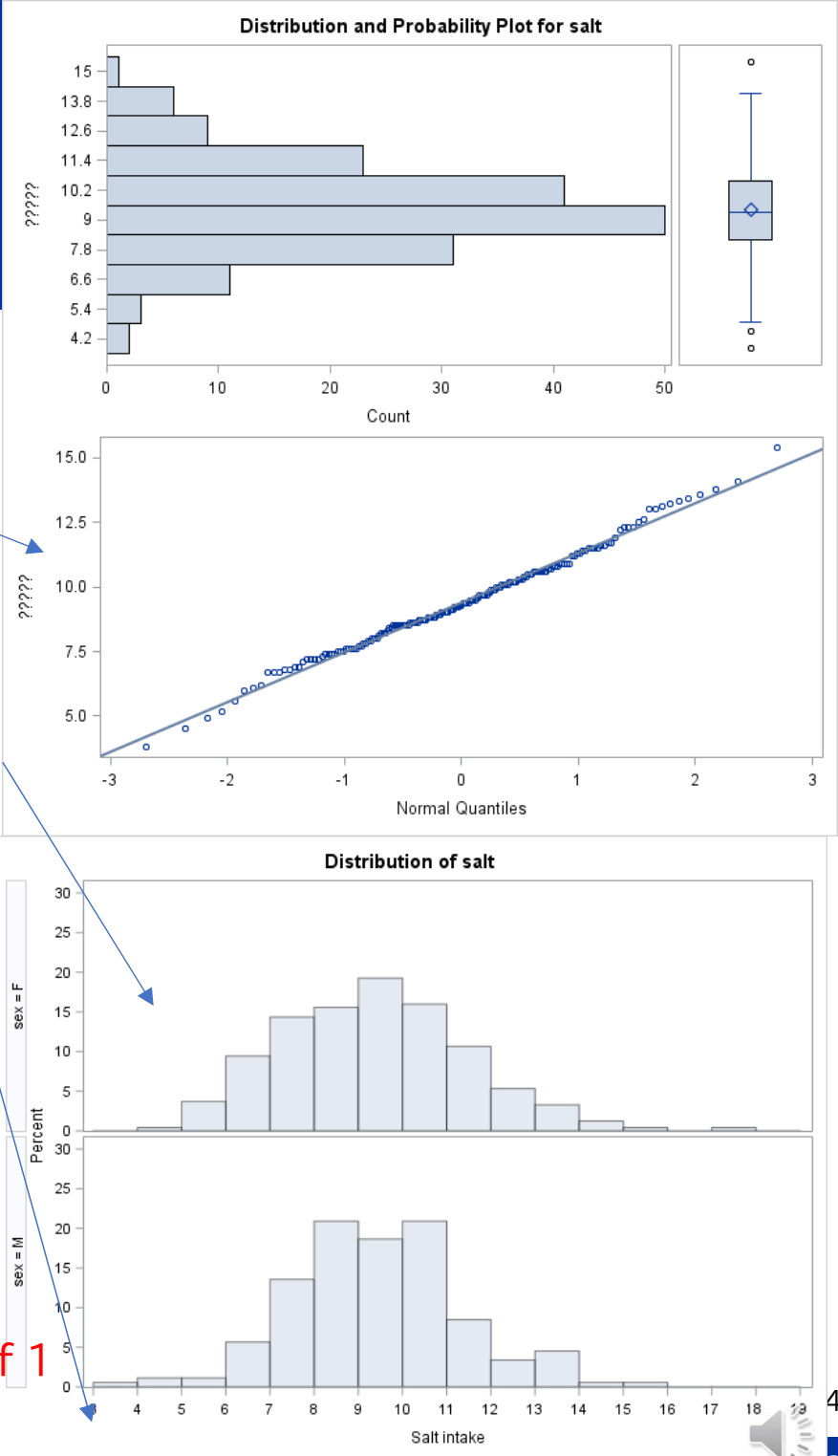
PROC univariate

⑤ histogram

```
proc univariate data=health plot normal;
```

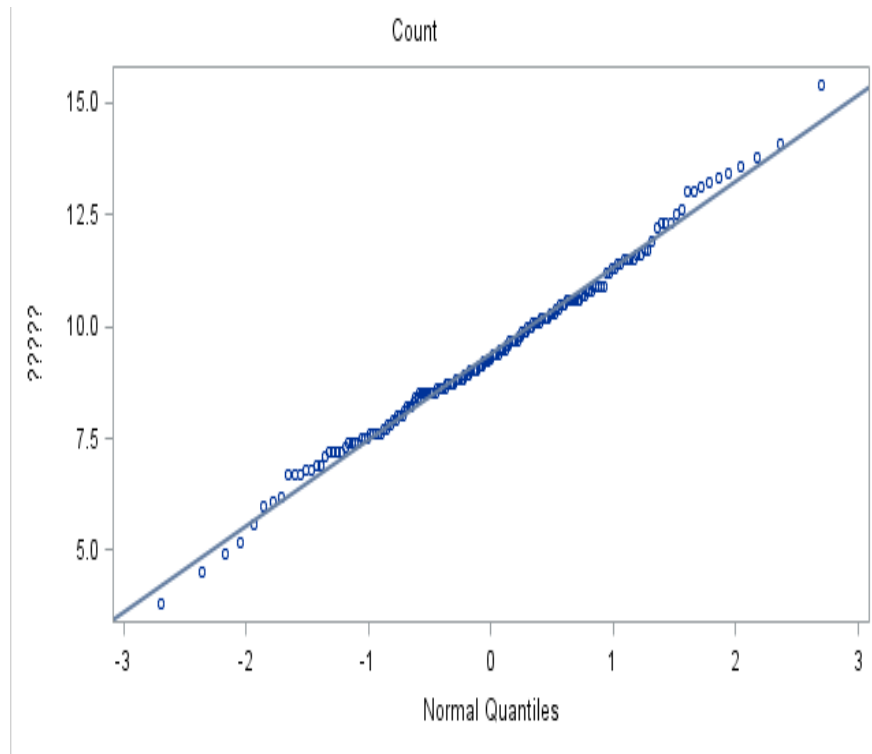
```
var salt;  
class sex;  
histogram salt/ endpoints=3 to 19 by 1;  
id id;  
label salt='Salt intake';  
run;
```

Scales from 3 to 19, with intervals of 1



PROC univariate

⑥ Normal probability plot (Q-Qplot) 正規確率プロット

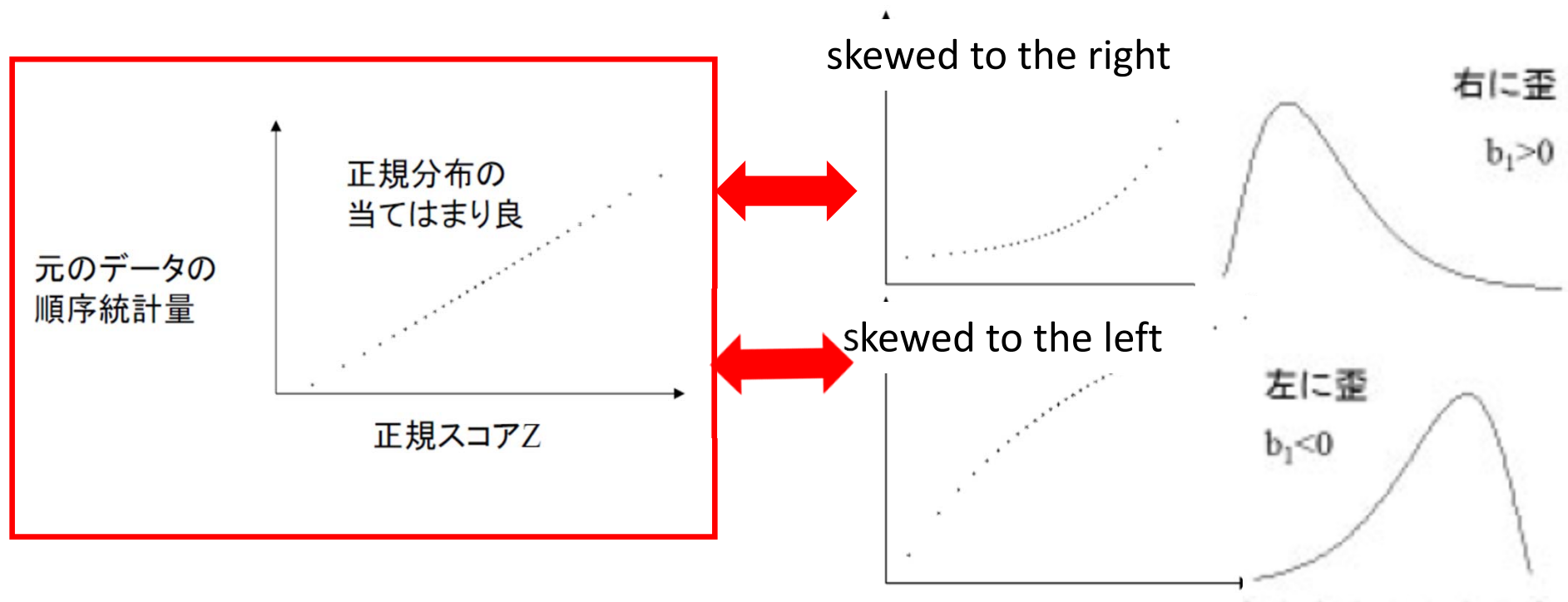


- データセットがほぼ正規分布しているかどうかを評価するためのグラフィカルな手法
- 正規確率プロットは次のように構成される。
 - 縦軸 順序付けられた応答値
 - 横軸: 正規順序統計量の中央値

PROC univariate

⑥ Normal probability plot

1. 正規確率プロットにより、データは正規分布しているか？ 判断します
2. 正規性からの逸脱はどのようなものか（データが歪んでいる、予想よりテールが短い、予想よりテールが長い）？





Q3 ヒストグラム

- Q3-1
 - Create a histogram of salt intake for each sex.
- Q3-2
 - Create a histogram for each gram of salt.

Today's overview

1. Usage of Statistical Software (package)

- 1) Brief Introduction to SAS
- 2) Reading Data

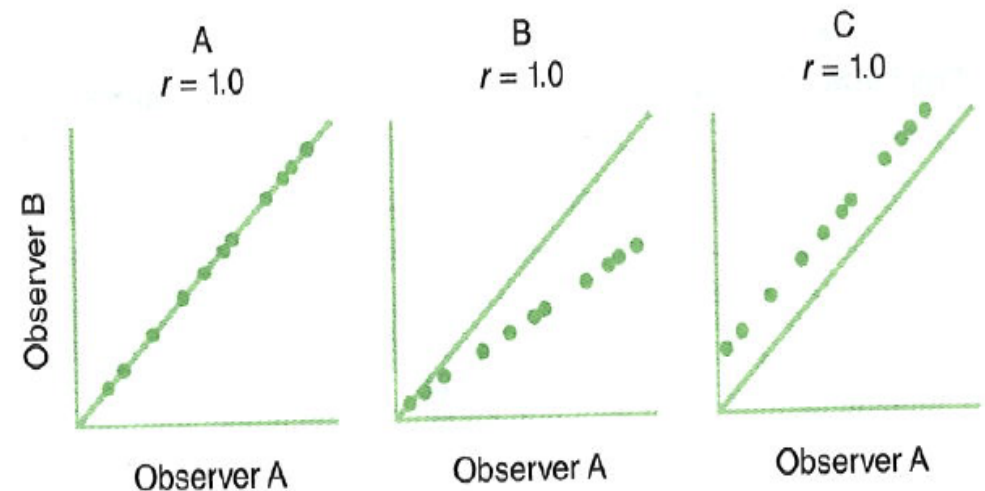
2. Description of Data

- 1) Location: mean, median, percentile
- 2) Dispersion: range, inter-quartile range, CV
- 3) Distribution: skewness, kurtosis, Normality
- 4) Association between two variables: Correlation

相関

- 相関は、2つの変数間の関連の強さ、線形の関連性を測定する二変量解析

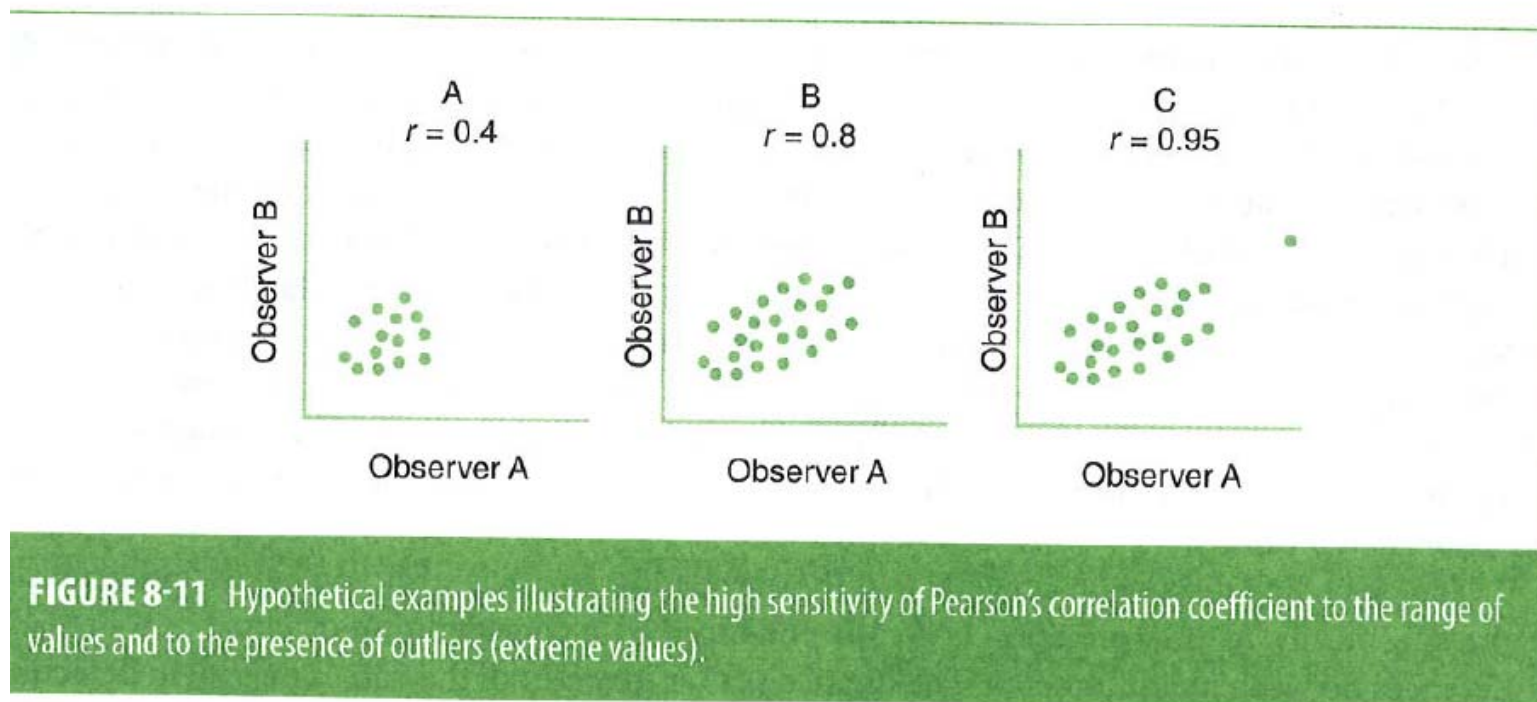
→ 相関係数は
のA,B,Cはすべて1.0である
→ 必ずしも一致の指標
とはならない。



- 相関係数の値は+1~-1の間で変化する。
- 相関係数の値が0に向かうほど、2つの変数の関係は弱くなる

Pearson product-moment correlation coefficient (usually denoted by “r”)

$$\frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}} = \text{Cov}(x, y) / \sqrt{V(x)V(y)}$$



Spearman correlation coefficient or rank correlation coefficient (denoted by “ ρ ”)

- 2つの変数間の関係の強さを測定するノンパラメトリックな検定。

$$\rho = 1 - \frac{6}{n^3 - n} \sum_{i=1}^n (R_i - R_i')^2$$

ρ = Spearman rank correlation

R_i, R_i' = rank

$R_i - R_i'$ = the difference between the ranks of corresponding variables

n = number of observations

- (x, y) を順位データ(1~ n)として変換し、ピアソン相関係数を計算する。
- 同値のデータがある場合は、平均順位として与えられる。

Kendall correlation coefficient (denoted “ τ (タウ)”)

- 2つの変数間の関連の強さを測定するノンパラメトリック検定
-
- aとbの2つの標本を考え、それぞれの標本サイズをnとすると、a bの組合せの総数は $n(n-1)/2$

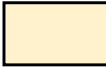
$$\tau = \frac{n_c - n_d}{\frac{1}{2}n(n-1)}$$

nc= Number of concordant
nd= Number of discordant

Ex. (x,y) = (1,1) (2,3) (3,5) (5,4)

(1,1)	$\leftrightarrow$	(2,3)	(3,5)	(5,4)
(2,3)			(3,5)	(5,4)
(3,5)				(5,4)

 . . . concordant
(normal order)

 . . . discordant
(reverse order)

$$\tau = \frac{5 - 1}{\frac{1}{2} \times 4 \times (4 - 1)} = \frac{4}{6} = \frac{2}{3}$$



#Q4 Use CORR 線形結合の強さを確認

```
proc corr option data=dataset name;
```



pearson, spearman, kendall

```
[var var.list;
```

```
[with var.list;
```

```
run;
```

Option

nomiss: Use only observations with no deficiencies
If not specified, all the observations that exist for each combination are used for the calculation.

Q4: health2データセットを用いて、血圧と食塩摂取量の相関係数を算出せよ。

```
/*corr procedure*/
```

```
proc corr pearson spearman kendall data=health2;
```

```
var sbp dbp salt;
```

```
run;
```

Correlation coefficients

Pearson

Pearson Correlation Coefficients, N = 421 Prob > r under H0: Rho=0			
	sbp	dbp	salt
sbp	1.00000	0.71894 <.0001	0.17764 0.0002
dbp	0.71894 <.0001	1.00000	0.15371 0.0016
salt	0.17764 0.0002	0.15371 0.0016	1.00000

Spearman

Spearman Correlation Coefficients, N = 421 Prob > r under H0: Rho=0			
	sbp	dbp	salt
sbp	1.00000	0.71932 <.0001	0.15706 0.0012
dbp	0.71932 <.0001	1.00000	0.14794 0.0023
salt	0.15706 0.0012	0.14794 0.0023	1.00000

Kendall

Kendall Tau b Correlation Coefficients, N = 421 Prob > tau under H0: Tau=0			
	sbp	dbp	salt
sbp	1.00000	0.53927 <.0001	0.10777 0.0012
dbp	0.53927 <.0001	1.00000	0.10160 0.0023
salt	0.10777 0.0012	0.10160 0.0023	1.00000

```

/*corr procedure*/;
proc corr pearson spearman
kendall data=health2;
  var sbp dbp salt;
run;
    
```

Output format for Matrix type ①

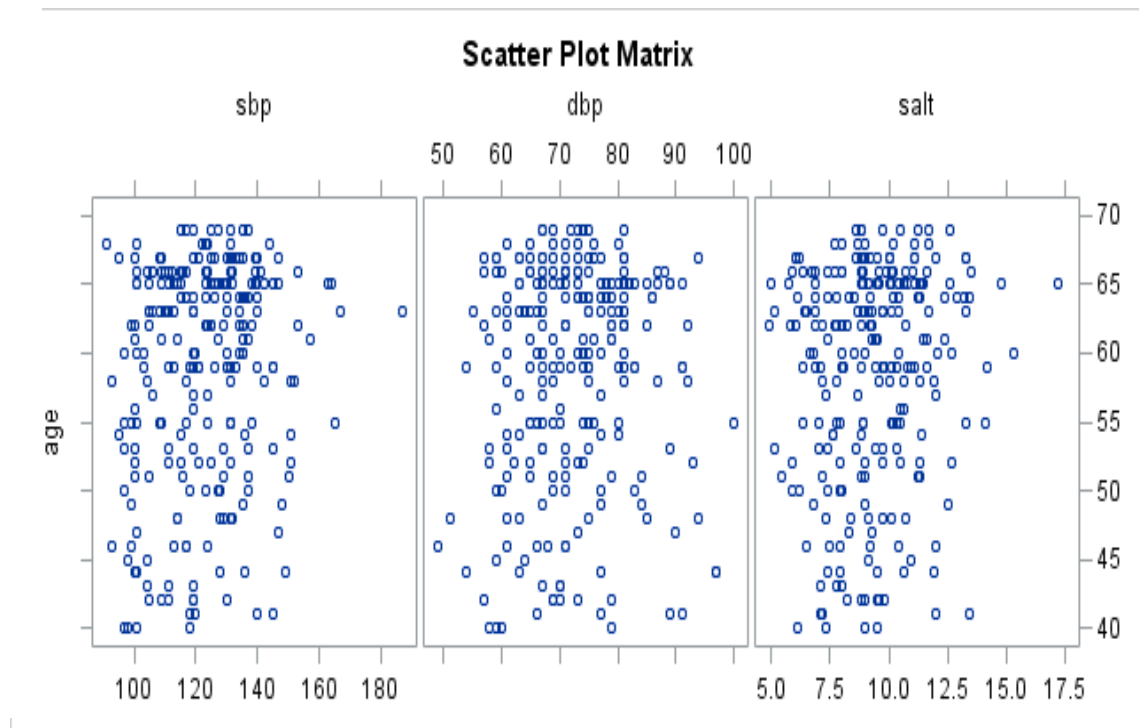
Simple Statistics						
Variable	N	Mean	Std Dev	Median	Minimum	Maximum
age	244	58.45492	8.05805	61.00000	40.00000	69.00000
sbp	244	122.72951	16.27189	123.00000	91.00000	187.00000
dbp	244	72.23361	9.50280	71.00000	49.00000	100.00000
salt	244	9.32541	2.10178	9.20000	4.90000	17.20000

Pearson Correlation Coefficients, N = 244 Prob > r under H0: Rho=0			
	sbp	dbp	salt
age	0.15953 0.0126	0.06836 0.2875	0.12452 0.0521

Spearman Correlation Coefficients, N = 244 Prob > r under H0: Rho=0			
	sbp	dbp	salt
age	0.14196 0.0266	0.07555 0.2397	0.12797 0.0458

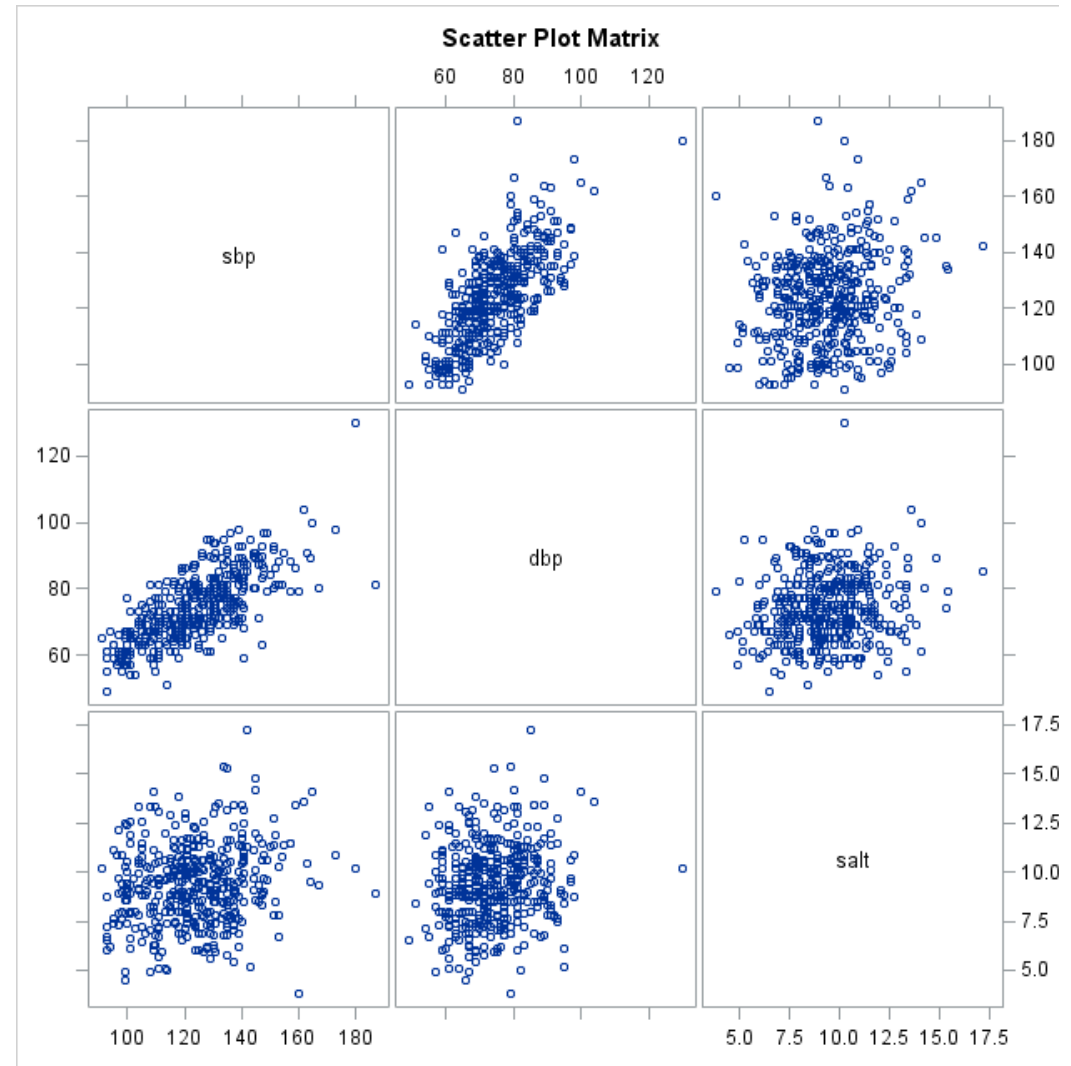
Kendall Tau b Correlation Coefficients, N = 244 Prob > tau under H0: Tau=0			
	sbp	dbp	salt
age	0.09794 0.0272	0.05322 0.2338	0.08532 0.0536

```
proc corr pearson spearman kendall
plots=matrix(histogram) data=health;
with age;
var sbp dbp salt;
by sex;
run;
```



Output format for Matrix type ②

```
*corr procedure;  
ods pdf file='F:¥corr.pdf';  
proc corr pearson  
spearman kendall  
plots=matrix  
data=health;  
    var sbp dbp salt;  
run;  
ods pdf close;  
run;
```



Output format for Matrix type ③

*corr procedure;

ods pdf

file='F:¥corr.pdf';

proc corr pearson

spearman kendall

plots=matrix(histogram)

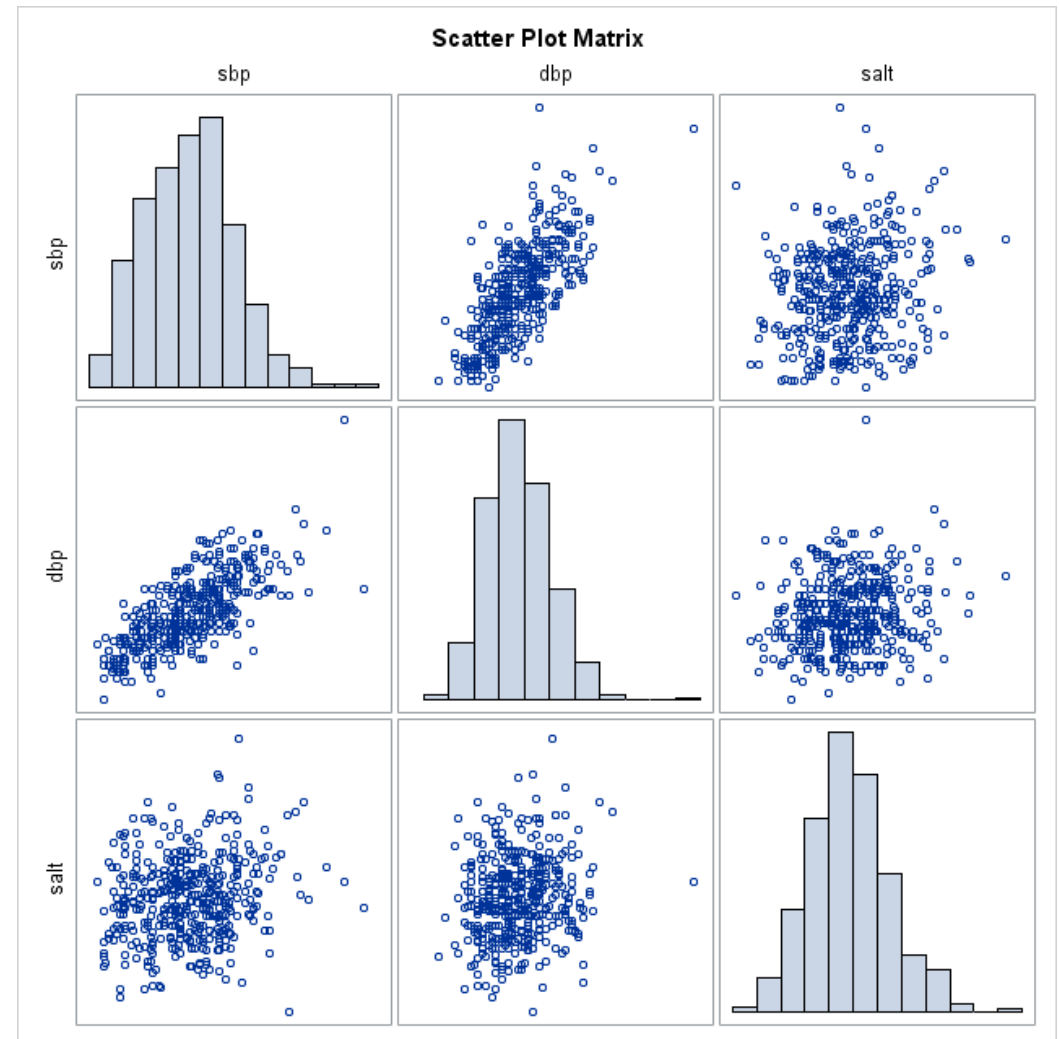
data=health;

var sbp dbp salt;

run;

ods pdf close;

run;



Polychoric / Polyserial Correlations

- Polychoric correlations (=tetrachoric correlation)
 - 2つの観測されたカテゴリ変数の間の相関。
- Polyserial Correlations
 - 連続変数とカテゴリカル変数の相関
- 血圧と食塩摂取量の相関
 - SBPと食塩摂取量
 - 高血圧カテゴリーと食塩摂取量
 - ピアソン
 - スピアマン
 - ポリシリアル(高血圧と食塩摂取量)

Polyserial correlation

```
proc corr data=health2 pearson polyserial;  
  with HTN_c;  
var age salt sbp;  
run;
```

“HTN_c”_is JSH hypertension grade as categorical variable

```
proc corr data=health2 pearson;  
  var age salt sbp;  
run;
```

Output Delivery System (ODS)

Excel output

```
ODS EXCEL FILE = 'file-specification' OPTIONS ( suboption(s) );
```

```
... Procedures ...
```

```
ODS EXCEL CLOSE;
```

ここで様々なオプションを指定することで、Excel
の出力をカスタマイズすることができます。

start_at='2,3'

Setting the output position (column, row)

sheet_name='MY SHEET'

Set (input) the sheet name.

sheet_interval= 'XXXXXX'

Set up how the sheets are divided.

Bygroup

BYグループごとにシートを分ける。

TABLE

プロシージャが出力する各テーブルごとにシートを分ける。

PROC

プロシージャで出力するテーブルごとにシートを区切る。

NONE

シートを区切らない(1枚のシートにすべてを出力する)

ODS Excel output

```
ods excel file='F:\¥corr.xlsx' OPTIONS( sheet_interval='proc' );  
proc corr pearson spearman kendall Plots=matrix data=health;  
    var sbp dbp salt;  
run;
```

```
proc corr data=health2 pearson spearman kendall  
plots=matrix(histogram) data=health;  
    var sbp dbp salt;  
run;
```

```
ods excel close;  
run;
```

ODS pdf出力

```
ods pdf file='F:¥corr.pdf';  
ods proclabel='matrix_corr_1';  
proc corr pearson spearman kendall  
plots=matrix data=health;  
  var sbp dbp salt;  
run;
```

```
ods proclabel='matrix_corr_2';  
proc corr data=health2 pearson spearman kendall  
plots=matrix(histogram) data=health;  
  var sbp dbp salt;  
run;  
ods pdf close;  
run;
```

Excel

Excel screenshot showing the CORR analysis results. The worksheet is titled "CORR プロシージャ".

変数: sbp dbp salt

単純統計量

変数	N	平均	標準偏差	中央値	最小値	最大値
sbp	421	123.88836	16.11578	123.00000	91.00000	187.00000
dbp	421	74.07601	10.24434	73.00000	49.00000	130.00000
salt	421	9.36200	2.02803	9.30000	3.80000	17.20000

Pearson の相関係数, N = 421
H0: Rho=0 に対する Prob > |r|

	sbp	dbp	salt
sbp	1.00000	0.71894 <.0001	0.17764 0.0002
dbp	0.71894 <.0001	1.00000	0.15371 0.0016
salt	0.17764 0.0002	0.15371 0.0016	1.00000

Spearman の相関係数, N = 421
H0: Rho=0 に対する Prob > |r|

	sbp	dbp	salt
sbp	1.00000	0.71932 <.0001	0.15706 0.0012
dbp	0.71932 <.0001	1.00000	0.14794 0.0023
salt	0.15706 0.0012	0.14794 0.0023	1.00000

Kendall の tau-b 偏相関係数, N = 421
H0: Tau=0 に対する Prob > |tau|

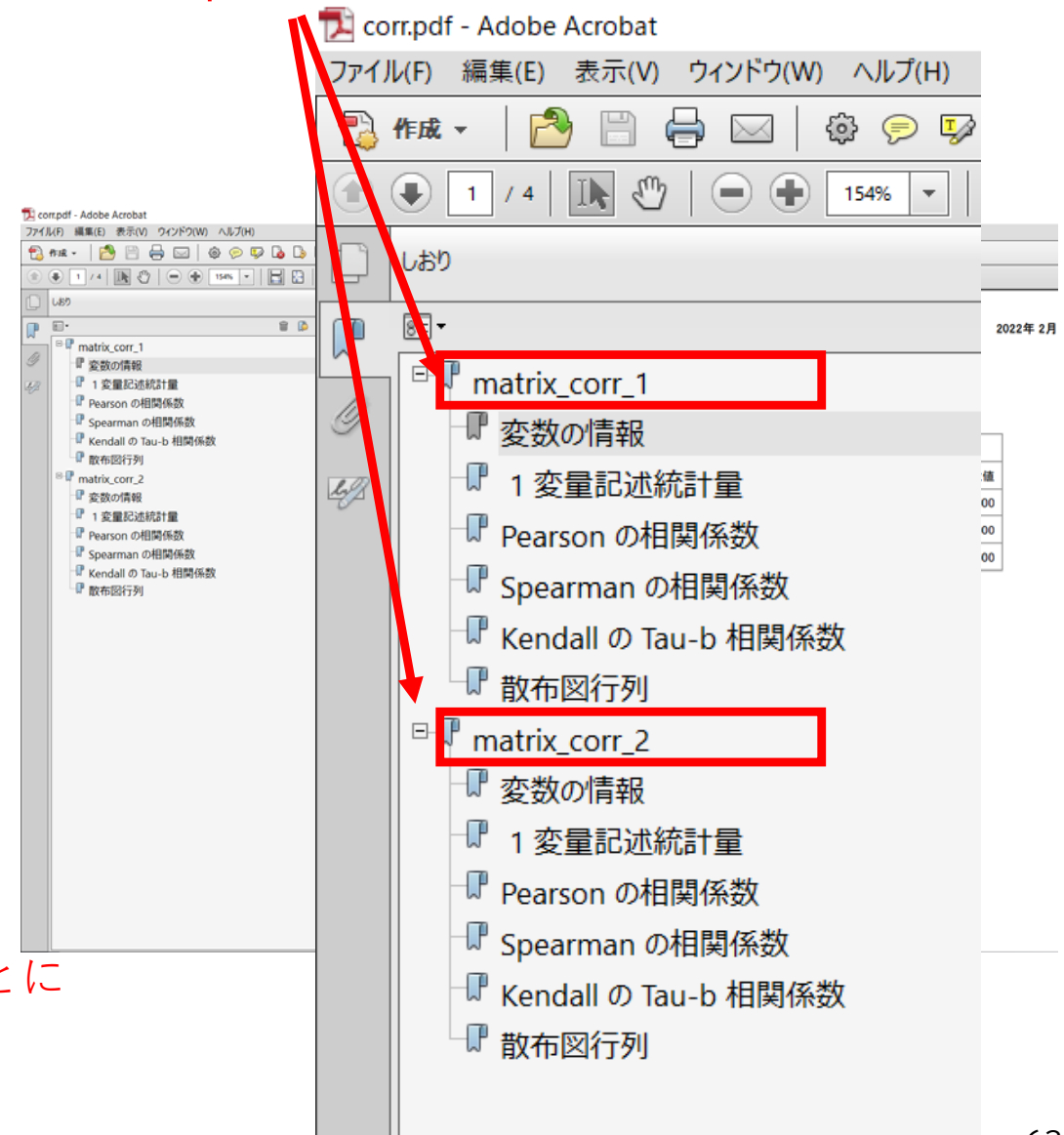
	sbp	dbp	salt
sbp	1.00000	0.53927 <.0001	0.10777 0.0012

手順で出力した表ごとにシートを分ける。

Proc 1 - Corr Proc 2 - Corr

PDF

Create a bookmark with a specified label



Summary

1. Usage of Statistical Software (package)

- 1) Brief Introduction to SAS
- 2) Reading Data

2. Description of Data

PROC Univariate

- 1) Location: mean, median, percentile
- 2) Dispersion: range, inter-quartile range, CV
- 3) Distribution: skewness, kurtosis, Normality

PROC CORR

- 4) Association between two variables: Correlation

Exercise

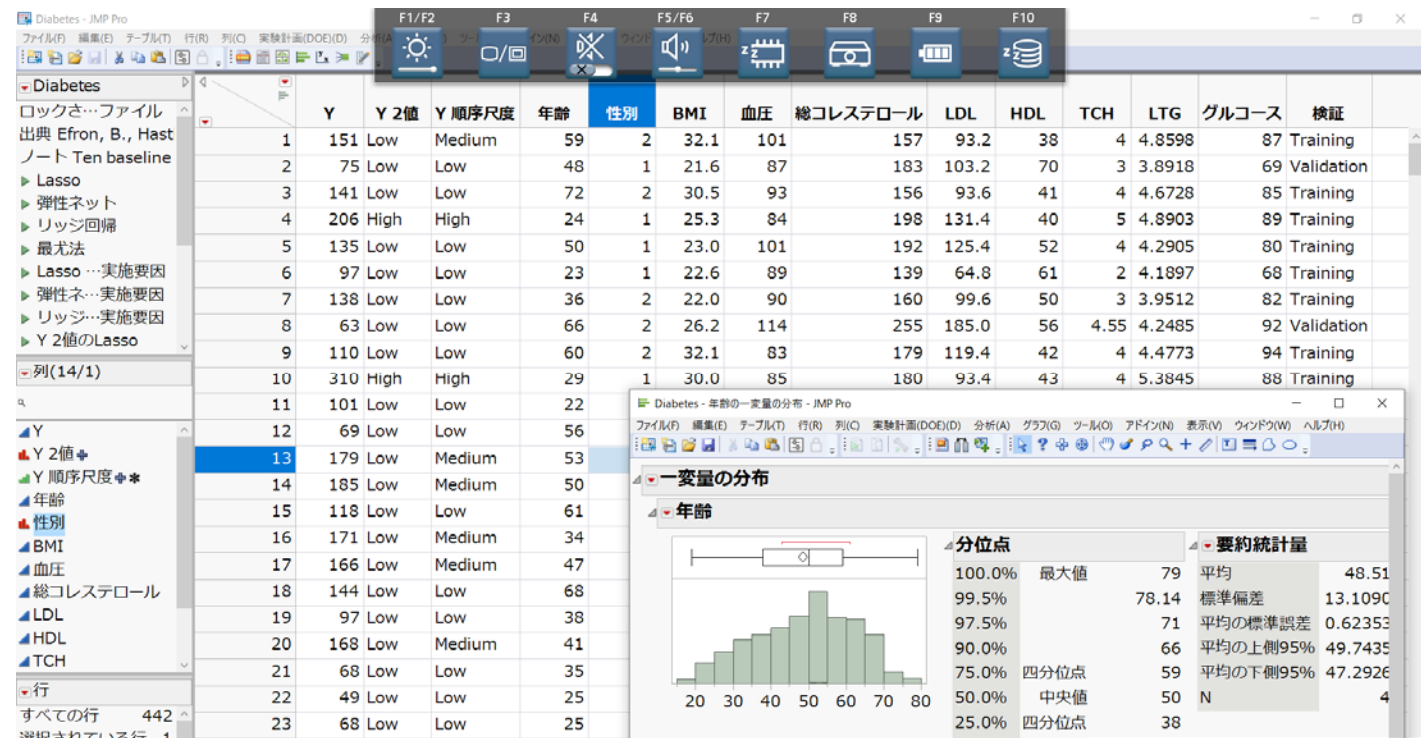
- 講義スライドに示された以下のQ1～Q4の課題について、統計パッケージを使って結果が再現できることを確認せよ。
 - Q1 データセット("health"→"health2")を作成する。
 - Q2 一変量手順を用いて、形状、位置、分布の指標を確認する(var= SBP, DBP)。
 - Q3 ヒストグラム
 - Q4 "health2 "データセットを用いて、血圧と食塩摂取量の相関係数を計算する。

R 他

以下は余裕があれば

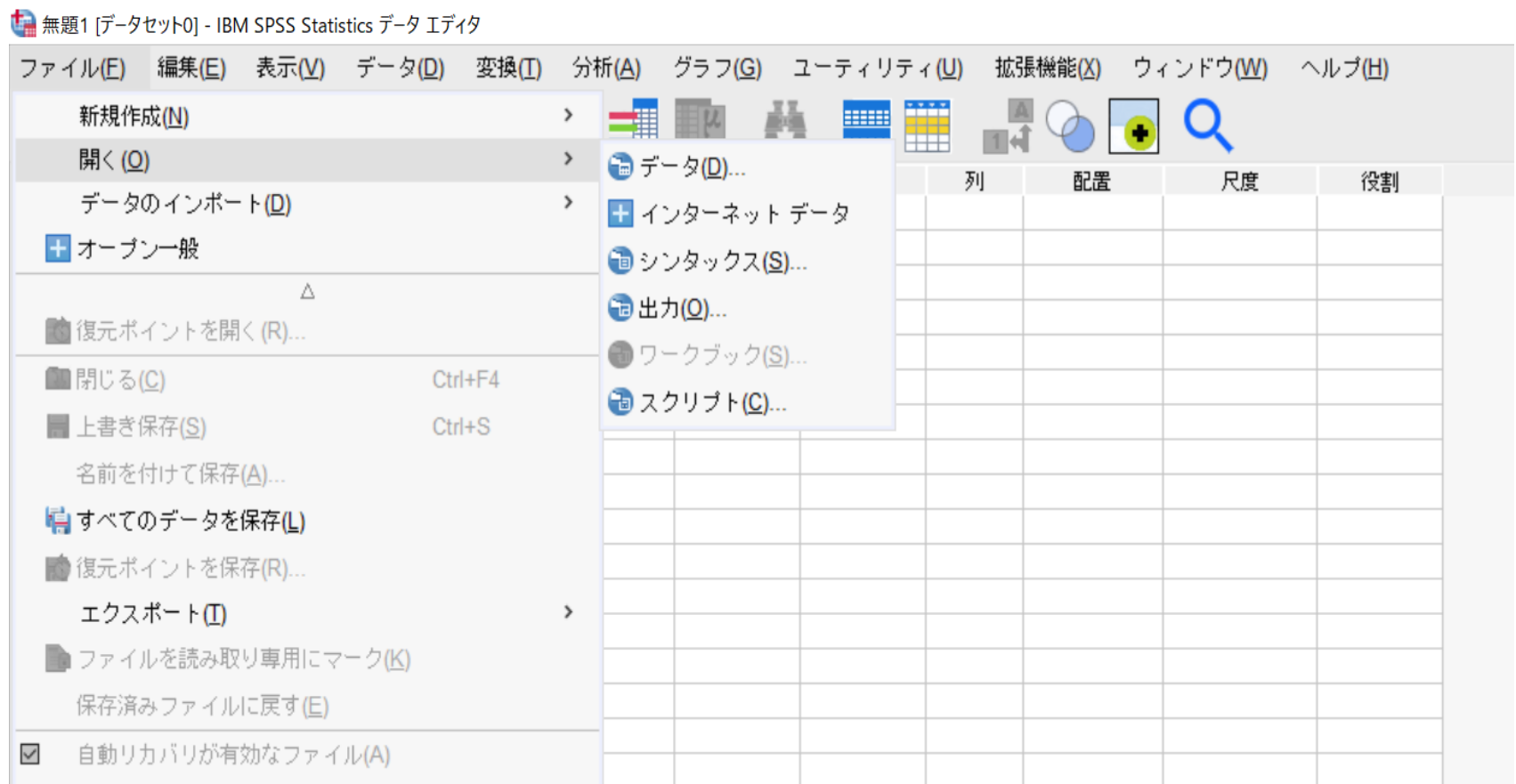
自分のPCでパッケージを使用するには JMP(JMP Pro) 製品版

- SAS社の製品
- 現在最新版は ver.16
- データの視覚化、クレンジング等の作業が行いやすい
- かなりの高度な解析までカバー
 - テキスト解析
 - 構造方程式 (SEM)などもカバー



自分のPCでパッケージを使用するには SPSS

- ・滋賀医科大学内ではMMCにてライセンス提供(使用可能)



自分のPCでパッケージを使用するには R studio

The screenshot displays the RStudio environment with the following components:

- Source Editor:** Contains the following R code:

```
1 install.packages("tidyverse")
2 library(tidyverse)
3 install.packages("tidyverse", dependencies=TRUE)
4 library(tidyverse)
```
- Console:** Shows the output of the R script, including the R version (4.1.2), copyright information, and the successful installation of the tidyverse package and its dependencies. It also lists the installed packages and their versions.

```
R version 4.1.2 (2021-11-01) -- "Bird Hippie"
Copyright (C) 2021 The R Foundation for Statistical Computing
Platform: x86_64-w64-mingw32/x64 (64-bit)

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

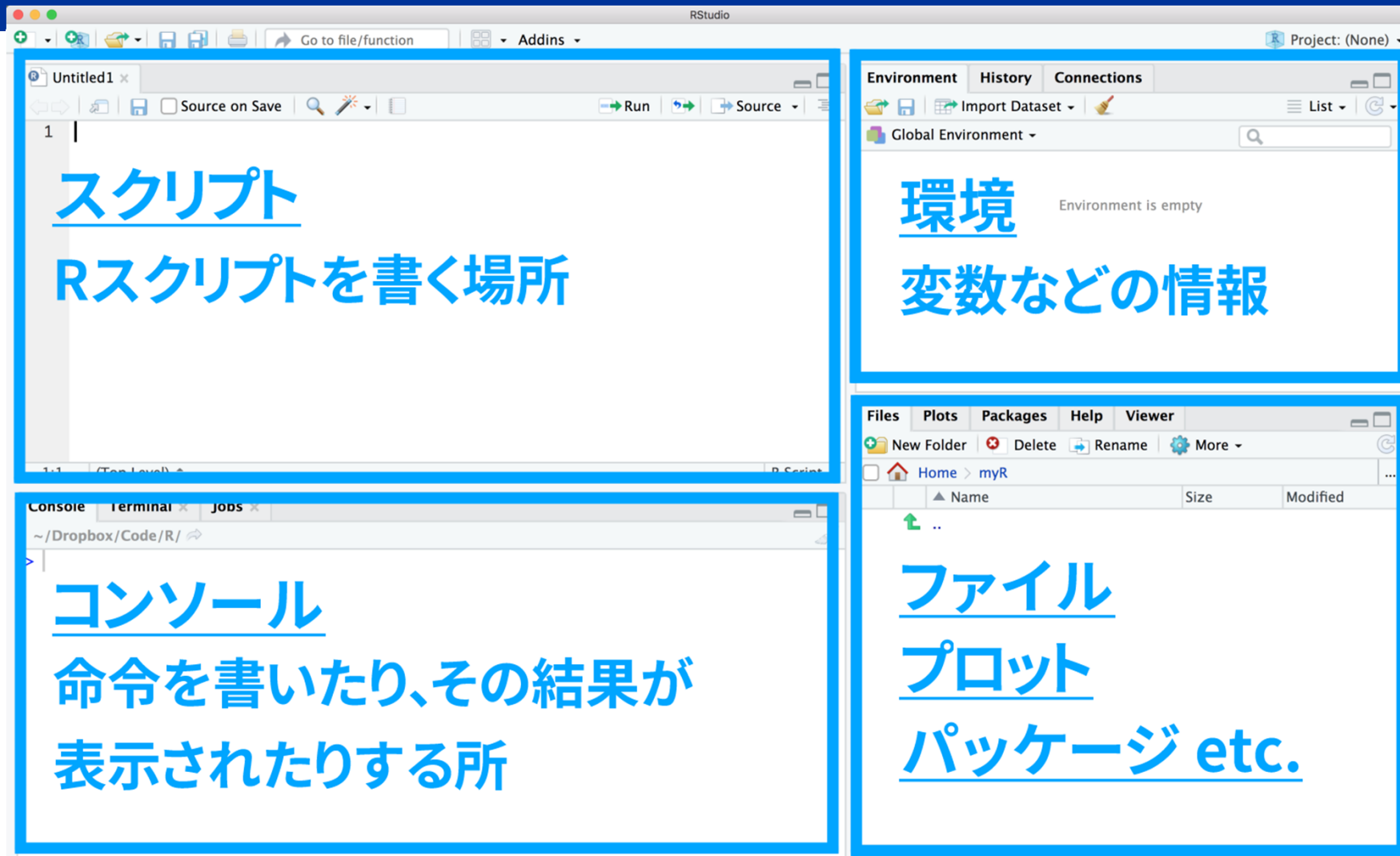
> install.packages("tidyverse", dependencies=TRUE)
WARNING: Rtools is required to build R packages but is not currently installed. Please download and install the appropriate version of Rtools before proceeding:
https://cran.rstudio.com/bin/windows/Rtools/
パッケージを 'c:/Users/HARADA/Documents/R/win-library/4.1' 中にインストールします
('lib' が指定されていないため)
trying URL 'https://cran.rstudio.com/bin/windows/contrib/4.1/tidyverse_1.3.1.zip'
Content type 'application/zip' length 430176 bytes (420 KB)
downloaded 420 KB

package 'tidyverse' successfully unpacked and MD5 sums checked

The downloaded binary packages are in
c:/Users/HARADA/AppData/Local/Temp/RtmpoimkG1/downloaded_packages
> library(tidyverse)
-- Attaching packages ----- tidyverse 1.3.1 --
✔ ggplot2 3.3.5    ✔ purrr 0.3.4
✔ tibble 3.1.5    ✔ dplyr 1.0.7
✔ tidyr 1.1.4     ✔ stringr 1.4.0
✔ readr 2.0.2     ✔ forcats 0.5.1
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag() masks stats::lag()
>
```
- Environment:** Shows the Global Environment with the message "Environment is empty".
- Files:** Shows the file explorer with the following files and their descriptions:

Name	Description	Version
mime	Map Filenames to MIME Types	0.12
modelr	Modelling Functions that Work with the Pipe	0.1.8
munsell	Utilities for Using Munsell Colours	0.5.0
openssl	Toolkit for Encryption, Signatures and Certificates Based on OpenSSL	1.4.5
pillar	Coloured Formatting for Columns	1.6.4
pkgconfig	Private Configuration for R Packages	2.0.3
pkgload	Simulate Package Installation and Attach	1.2.3
praise	Praise Users	1.0.0
prettyunits	Pretty, Human Readable Formatting of Quantities	1.1.1
processx	Execute and Control System Processes	3.5.2
progress	Terminal Progress Bars	1.2.2
ps	List, Query, Manipulate System Processes	1.6.0
purrr	Functional Programming Tools	0.3.4
R6	Encapsulated Classes with Reference Semantics	2.5.1
rappdirs	Application Directories: Determine Where to Save Data, Caches, and Logs	0.3.3
RColorBrewer	ColorBrewer Palettes	1.1-2
Rcpp	Seamless R and C++ Integration	1.0.7
readr	Read Rectangular Text Data	2.0.2
readxl	Read Excel Files	1.3.1
rematch	Match Regular Expressions with a Nicier API	1.0.1
rematch2	Tidy Output from Regular Expression Matching	2.1.2
reprex	Prepare Reproducible Example Code via the Clipboard	2.0.1
rex	Friendly Regular Expressions	1.2.0
rlang	Functions for Base Types and Core R and Tidyverse Features	0.4.12
rmarkdown	Dynamic Documents for R	2.11
rprojroot	Finding Files in Project Subdirectories	2.0.2
rstudioapi	Safely Access the RStudio API	0.13
rvest	Easily Harvest (Scrape) Web Pages	1.0.2

R studio



RStudioは、Rを使用するための開発環境（プログラミングのためのアプリケーション）です。Rを用いた統計解析は、R単体でも（RStudioを使わなくても）可能ですが、RStudioを使って統計解析を行う人が多い。

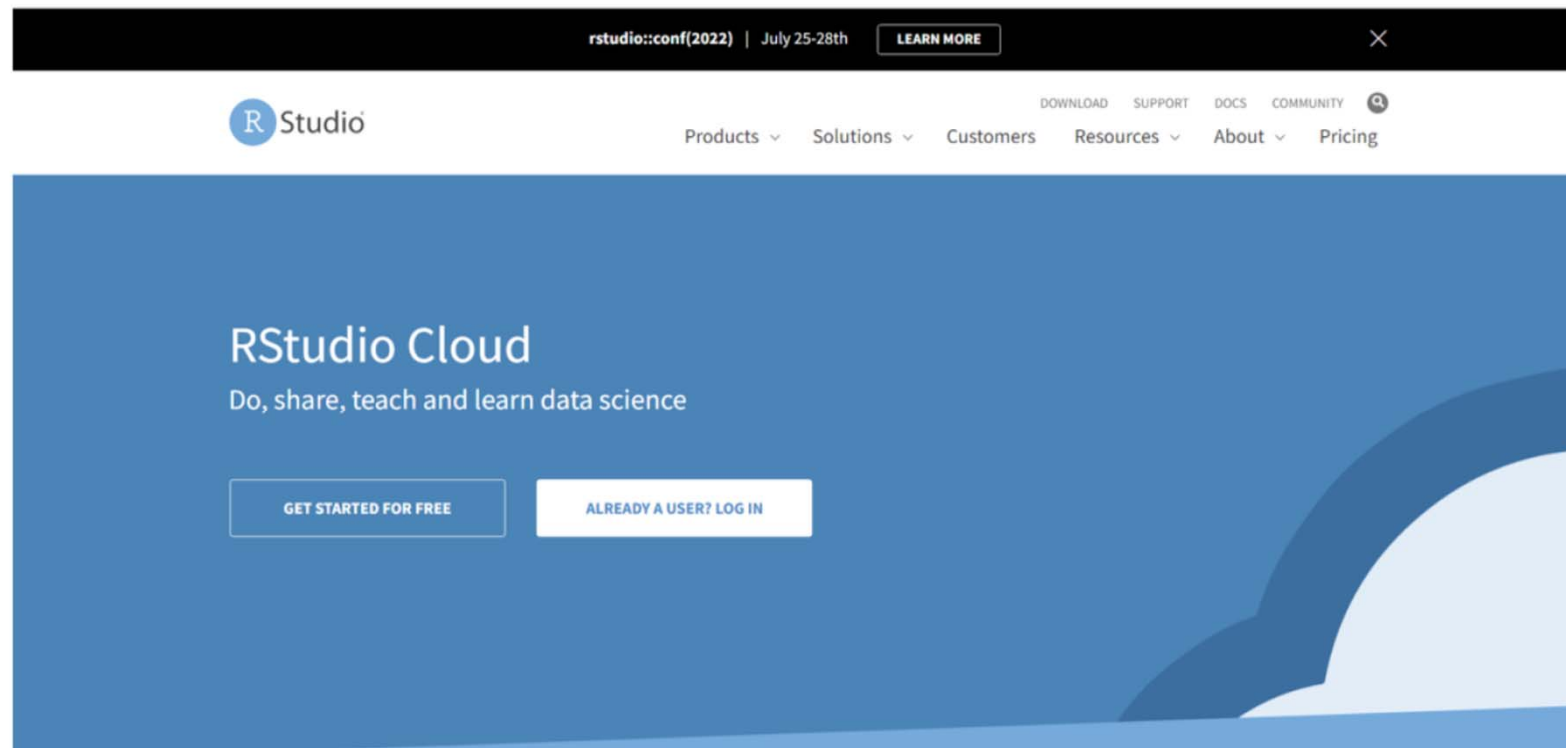
R-studio Cloud版

RStudio Cloud Introduction memo

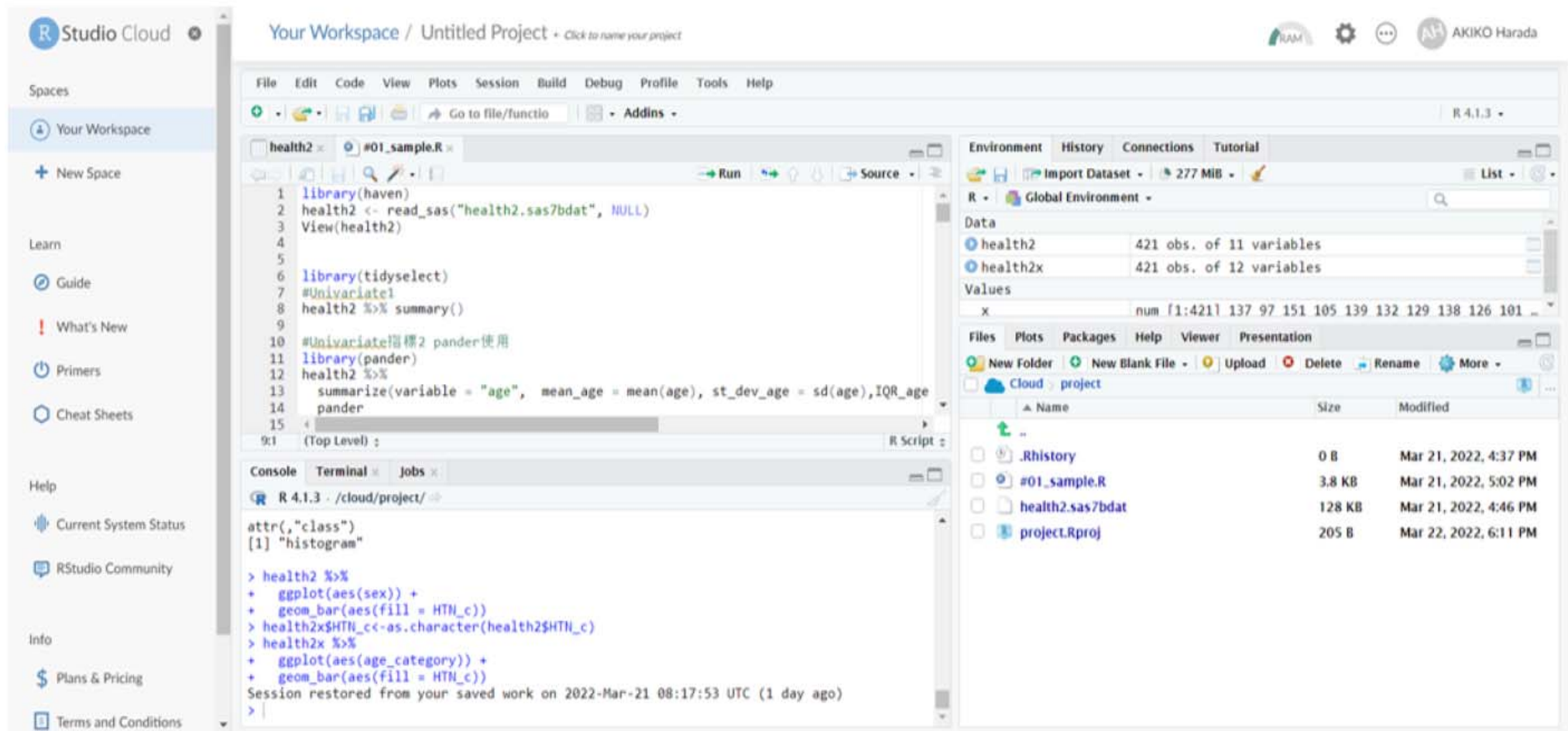
1 Register as a user (ユーザ登録を行う)

RStudio Cloud <https://www.rstudio.com/products/cloud/> (<https://www.rstudio.com/products/cloud/>)

Rソフトを自身のPCにインストールして使用してもよいが、クラウド版は、
・導入までの手間が回避
・いつでもどこでも最新環境が使用できるなどのメリットもある



Rstudio cloud screen



RSTUDIO Cloud screen

データセットのアップロード

3 Upload dataset

To upload datasets, click on the Upload button in environment window. Click the "File selection button", select the file to upload, and click "OK".

Upload Files

Target directory:
/cloud/project Browse...

File to upload:
ファイルを選択 lab.sas7bdat

TIP: To upload multiple files or a directory, create a zip file. The zip file will be automatically expanded after upload.

OK Cancel

Select and upload SAS file, csv, etc.

データセットの読み込み

The screenshot illustrates the steps to import a SAS dataset into RStudio Cloud. The interface shows the 'Environment' pane with 'Global Environment' selected. The 'Files' pane on the left lists project files, including 'lab.sas7bdat'. The 'Import Dataset' button in the top toolbar is highlighted. The 'Import Statistical Data' dialog box is open, showing the file path '/cloud/project/lab.sas7bdat' and a preview of the data. The 'Import' button is highlighted.

age	weight	h_bp	l_bp	tp	got	gpt	gspt	bun	age10
43	61.0	147.5	110	78	7.6	19	14	4.4	4
82	56.0	144.3	158	70	7.6	27	20	5.9	8
40	46.5	146.0	110	76	7.3	14	12	3.0	4
71	57.5	146.1	186	118	8.5	22	14	7.5	7
50	58.0	156.5	160	98	7.3	15	13	5.9	5
53	53.5	144.8	150	90	7.6	25	20	6.9	5

必要なパッケージの読み込み、コードの入力 などを行いながら解析

4 Enter code on the Source screen to start analysis.

Summary statistics using “Lab” dataset

The screenshot displays the RStudio interface with two panels. The left panel shows the 'Source' editor with the following code:

```
1 library(haven)
2 lab <- read_sas("lab.sas7bdat", NULL)
3
4 summary(lab)
```

A red arrow points from the code in the left panel to the right panel. The right panel shows the 'Console' output, which includes the summary statistics for the 'lab' dataset. The output is as follows:

```
R 4.1.3 . /cloud/project/
> health2 %>%
+   ggplot(aes(sex)) +
+   geom_bar(aes(fill = HTN_c))
> health2$HTN_c<-as.character(health2$HTN_c)
> health2x %>%
+   ggplot(aes(age_category)) +
+   geom_bar(aes(fill = HTN_c))
Session restored from your saved work on 2022-Mar-21 08:17:53 UTC (1 day ago)
> library(haven)
> lab <- read_sas("lab.sas7bdat", NULL)
> View(lab)
> View(lab)
> summary(lab)
```

age		weight		h_bp		l_bp		tp	
Min.	:40.00	Min.	:40.00	Min.	:136.5	Min.	:108.0	Min.	: 70.00
1st Qu.	:49.00	1st Qu.	:48.25	1st Qu.	:144.6	1st Qu.	:112.0	1st Qu.	: 73.00
Median	:54.00	Median	:50.00	Median	:146.1	Median	:142.0	Median	: 78.00
Mean	:58.27	Mean	:51.64	Mean	:147.1	Mean	:139.3	Mean	: 84.91
3rd Qu.	:66.50	3rd Qu.	:56.75	3rd Qu.	:151.1	3rd Qu.	:159.0	3rd Qu.	: 94.00
Max.	:82.00	Max.	:61.00	Max.	:156.5	Max.	:186.0	Max.	:118.00

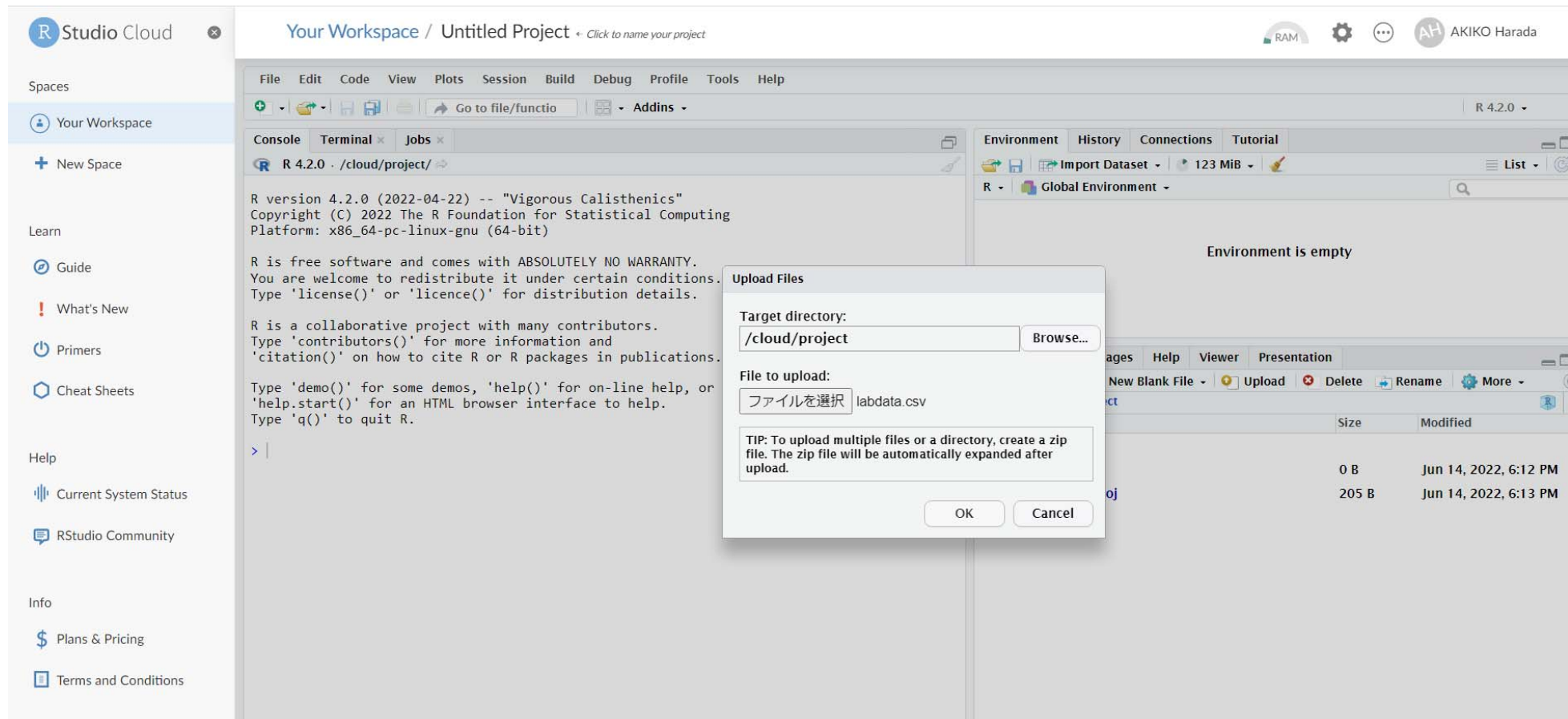
The right panel also shows a table of summary statistics for the 'lab' dataset, which is highlighted with a red box. This table provides a detailed summary of the data, including minimum, first quartile, median, mean, third quartile, and maximum values for various variables.

講義の例題をRで確認してみる

•「helath2.csv」

The screenshot displays the RStudio Cloud interface. The top bar shows 'Your Workspace / Untitled Project' and the user 'AKIKO Harada'. The left sidebar contains navigation links for Spaces, Your Workspace, New Space, Learn (Guide, What's New, Primers, Cheat Sheets), and Help (Current System Status, RStudio Community, Info, Plans & Pricing, Terms and Conditions). The main console area shows the R version 4.2.0 (2022-04-22) and the path '/cloud/project/'. The environment pane on the right indicates 'Environment is empty'. The file explorer at the bottom right shows a list of files: '..' (0 B, Jun 14, 2022, 6:12 PM), '.Rhistory' (0 B, Jun 14, 2022, 6:12 PM), and 'project.Rproj' (205 B, Jun 14, 2022, 6:13 PM). The 'Upload' button in the file explorer is circled in red.

Health2.csvをアップロード



The screenshot displays the RStudio Cloud interface. On the left, the 'Spaces' sidebar shows 'Your Workspace' and 'New Space'. The main area is titled 'Your Workspace / Untitled Project'. The console pane shows the R version 4.2.0 and its license. The environment pane is empty. An 'Upload Files' dialog box is open in the center, with the following details:

- Target directory:
- File to upload: labdata.csv
- TIP: To upload multiple files or a directory, create a zip file. The zip file will be automatically expanded after upload.
- Buttons:

Import datasetから CSV形式は「Text(base)」

The screenshot displays the RStudio Cloud interface. The top bar shows 'Your Workspace / Untitled Project' and 'R 4.2.0'. The left sidebar contains navigation options like 'Spaces', 'Your Workspace', 'New Space', 'Learn', and 'Help'. The main console area shows the R version 4.2.0 startup message. The 'Environment' pane on the right shows 'Import Dataset' with a dropdown menu open, listing options: 'From Text (base)...', 'From Text (readr)...', 'From Excel...', 'From SPSS...', 'From SAS...', and 'From Stata...'. A large red arrow points to 'From Text (base)...'. Below the menu, a file selection dialog is open, showing a list of files in the 'cloud > project' directory: '.Rhistory' (0 B), 'project.Rproj' (205 B), and 'labdata.csv' (546 B). The 'labdata.csv' file is selected.

Name	Size	Modified
..		
.Rhistory	0 B	Jun 14, 2022, 6:12 PM
project.Rproj	205 B	Jun 14, 2022, 6:13 PM
labdata.csv	546 B	Jun 14, 2022, 6:16 PM

Import Dataset

Name: labdata

Encoding: Platform

Heading: ☒ Yes ☐ No

Row names: Automatic

Separator: Comma

Decimal: Period

Quote: Double (")

Comment: None

na.strings: NA

☐ Strings as factors

Input File

ID, sex, age, weight, height, h_bp, l_bp, tp, got, gpt, alp, ggpt, bun
 1, F, 43, 61.0, 147.5, 110, 78, 7.6, 19, 14, 4.4, 10, 11
 2, F, 82, 56.0, 144.3, 158, 70, 7.6, 27, 20, 5.9, 7, 15
 3, M, 40, 46.5, 146.0, 110, 76, 7.3, 14, 12, 3.0, 8, 16
 4, M, 71, 57.5, 146.1, 186, 118, 8.5, 22, 14, 7.5, 22, 15
 5, F, 50, 58.0, 156.5, 160, 98, 7.3, 15, 13, 5.9, 8, 14
 6, F, 53, 53.5, 144.8, 150, 90, 7.6, 25, 20, 6.9, 12, 13
 7, M, 54, 50.0, 139.3, 164, 100, 7.1, 20, 17, 4.2, 23, 18
 8, M, 76, 40.0, 136.5, 130, 70, 7.7, 24, 16, 4.7, 12, 19
 9, M, 62, 48.0, 151.9, 114, 76, 6.8, 20, 17, 5.4, 10, 14
 10, F, 62, 49.0, 154.6, 142, 88, 8.0, 24, 17, 5.5, 16, 13
 11, M, 48, 48.5, 150.2, 108, 70, 7.4, 14, 14, 4.7, 9, 12

Data Frame

ID	sex	age	weight	height	h_bp	l_bp	tp	got	gpt	alp
1	F	43	61.0	147.5	110	78	7.6	19	14	4.4
2	F	82	56.0	144.3	158	70	7.6	27	20	5.9
3	M	40	46.5	146.0	110	76	7.3	14	12	3.0
4	M	71	57.5	146.1	186	118	8.5	22	14	7.5
5	F	50	58.0	156.5	160	98	7.3	15	13	5.9
6	F	53	53.5	144.8	150	90	7.6	25	20	6.9
7	M	54	50.0	139.3	164	100	7.1	20	17	4.2
8	M	76	40.0	136.5	130	70	7.7	24	16	4.7
9	M	62	48.0	151.9	114	76	6.8	20	17	5.4
10	F	62	49.0	154.6	142	88	8.0	24	17	5.5
11	M	48	48.5	150.2	108	70	7.4	14	14	4.7

Import Cancel

IMPORTする
 ファイルの1行目が変数見出しの
 場合などは”Heading”をYesと
 する

Environment

Global Environment

labdata 11 obs. of 13 variables

Files

Name	Size	Modified
labdata.csv	546 B	Jun 14, 2022, 6:16 PM

コードを入力するため、New fileから「R Script」を選択

The screenshot displays the RStudio Cloud web interface. The top bar shows 'Your Workspace / Untitled Project' and the user 'AKIKO Harada'. The left sidebar contains navigation links for Spaces, Learn, Help, and Info. The main editor area has a menu bar with 'File', 'Edit', 'Code', 'View', 'Plots', 'Session', 'Build', 'Debug', 'Profile', 'Tools', and 'Help'. The 'File' menu is open, showing options like 'New File', 'Open File...', 'Recent Files', 'Import Dataset', 'Save', 'Print...', 'Close', and 'Close All'. The 'New File' submenu is also open, listing various file types including 'R Script' (highlighted), 'Quarto Document...', 'R Notebook', 'R Markdown...', 'Shiny Web App...', 'Plumber API...', 'C File', 'C++ File', 'Header File', 'Markdown File', 'HTML File', 'CSS File', 'JavaScript File', 'D3 Script', 'Python Script', 'Shell Script', 'SQL Script', 'Stan File', 'Text File', 'R Sweave', 'R HTML', and 'R Documentation...'. The background editor shows a data table with columns 'alp', 'ggpt', and 'bun'. The right-hand pane includes tabs for 'Environment', 'History', 'Connections', and 'Tutorial', and a 'Files' section showing a file explorer view.

	alp	ggpt	bun
4	4.4	10	11
0	5.9	7	15
2	3.0	8	16
4	7.5	22	15
3	5.9	8	14
0	6.9	12	13
7	4.2	23	18
6	4.7	12	19
7	5.4	10	14
7	5.5	16	13
4	4.7	9	12

データの記述

tableone, tidyverseパッケージ等を使用

- データの記述の例題として「**healthc.csv**」を用意
- 健診データ(仮想)で、
 - ID, 性別, 年齢, 収縮期血圧(SBP)、拡張期血圧(DBP)、推定食塩摂取量(SALT)、高血圧分類(HTN_c : 1-7) などからなる

至適高血圧	sbp<120 and dbp< 80 HTN_c= 1
正常高血圧	120<=sbp<130 and dbp< 80 HTN_c=2;
正常高値血圧	130<=sbp<140 or 80<=dbp< 90 HTN_c=3;
I 度高血圧	140<=sbp<160 or 90<=dbp< 100 HTN_c=4;
II 度高血圧	160<=sbp<180 or 100<=dbp< 110 HTN_c=5;
III 度高血圧	180<=sbp and 110<=dbp HTN_c=6;
Other HTN	=_c=7;

- データの記述→記述統計→(血圧データについて)散布図→箱ひげ図→ヒストグラム→ヴァイオリンプロットのコード例を示してある

Data import

○Lab.sas

```
library(haven)
```

```
lab2 <- read_sas("D:/lab2.sas7bdat", NULL)
```

```
View(lab2)
```

○Health2.sas

```
library(haven)
```

```
health2 <- read_sas("D:/health2.sas7bdat", NULL)
```

```
View(health2)
```

Shuirup #X q lyduldw h#D q do| vlv#lq #J

#Univariate指標1 pander使用

```
> health2 %>%
```

```
summarize (variable="age", mean_age = mean(age),  
st_dev_age = sd(age), IQR_age = IQR(age), var_age =  
var (age)) %>%
```

```
pander()
```

```
-----  
mean_age  st_dev_age  IQR_age  var_age
```

```
-----  
58.9      7.816      11      61.09
```

#Univariate指標2

health2 %>% summary()

```
Console Terminal x Jobs x
R 4.1.2 · F:/biostat_ex/
> summarize(mean_age = mean(age), st_dev_age = sd(age), IQR_age = IQR(age), var_age = var (age))
Error: unexpected '=' in "summarize(mean_age = mean(age), st_dev_age = sd(age), IQR=age ="
> library(dplyr)
> summarize(mean_age = mean(age), st_dev_age = sd(age), IQR_age = IQR(age), var_age = var (age))
# A tibble: 1 x 4
  mean_age st_dev_age IQR_age var_age
  <dbl>    <dbl>    <dbl>    <dbl>
1    58.9      7.82      11      61.1

> health2 %>% summary()
      sex      age      sbp      dbp      salt      id      sbp_c
Length:421   Min. :40.0   Min. : 91.0   Min. : 49.00   Min. : 3.800   Min. : 1   Min. :1.000
Class :character 1st Qu.:54.0   1st Qu.:112.0  1st Qu.: 67.00   1st Qu.: 7.900   1st Qu.:106   1st Qu.:1.000
Mode :character  Median :61.0   Median :123.0  Median : 73.00   Median : 9.300   Median :211   Median :2.000
              Mean :58.9   Mean :123.9   Mean : 74.08   Mean : 9.362   Mean :211   Mean :2.172
              3rd Qu.:65.0   3rd Qu.:135.0  3rd Qu.: 81.00   3rd Qu.:10.600   3rd Qu.:316   3rd Qu.:3.000
              Max. :69.0   Max. :187.0   Max. :130.00   Max. :17.200   Max. :421   Max. :7.000
              NA's :2
```

#psych package

vars	n	mean	sd		median	trimmed	mad	min	max
X1	1	421	123.89	16.12	123	123.44	17.79	91	187
		range	skew	kurtosis	se				
X1	96	0.36	0.29	0.79					

#tableone

- Create summary tables like background information

```
library(tableone)
```

```
tbl_1 <- CreateTableOne(vars = c("sex", "age", "sbp",  
"dbp", "salt", "HTN_c"), strata="age_category", factorVars  
= c("sex", "HTN_c"), data = health2x)
```

Vars:

Variable names (columns) to be included in the table

Strata:

Stratified Variables

factorVars:

Categorical variable out of vars

data: Data set name

```
tbl_1
```

```
tbl_1 %>%
```

```
print() %>%
```

```
write.csv(file = "tableone.csv")
```

Stratified by age_category						
	64	40s	50s	60s	p	test
n	64	113	244			
sex = M (%)	23 (35.9)	43 (38.1)	111 (45.5)			0.234
age (mean (SD))	44.42 (2.90)	55.12 (2.94)	64.45 (2.57)			<0.001
sbp (mean (SD))	117.70 (15.04)	123.86 (16.15)	125.52 (16.04)			0.002
dbp (mean (SD))	73.05 (11.85)	74.75 (10.47)	74.03 (9.69)			0.566
salt (mean (SD))	9.02 (1.66)	9.35 (1.92)	9.46 (2.16)			0.312
HTN_c (%)						0.087
1	36 (56.2)	45 (39.8)	80 (32.8)			
2	9 (14.1)	18 (15.9)	47 (19.3)			
3	15 (23.4)	37 (32.7)	99 (40.6)			
4	4 (6.2)	12 (10.6)	15 (6.1)			
5	0 (0.0)	1 (0.9)	2 (0.8)			
6	0 (0.0)	0 (0.0)	1 (0.4)			

tableone.csv

	40s	50s	60s	p	test
n	64	113	244		
sex = M (%)	23 (35.9)	43 (38.1)	111 (45.5)	0.234	
age (mean (SD))	44.42 (2.90)	55.12 (2.94)	64.45 (2.57)	<0.001	
sbp (mean (SD))	117.70 (15.04)	123.86 (16.15)	125.52 (16.04)	0.002	
dbp (mean (SD))	73.05 (11.85)	74.75 (10.47)	74.03 (9.69)	0.566	
salt (mean (SD))	9.02 (1.66)	9.35 (1.92)	9.46 (2.16)	0.312	
HTN_c (%)				0.087	
1	36 (56.2)	45 (39.8)	80 (32.8)		
2	9 (14.1)	18 (15.9)	47 (19.3)		
3	15 (23.4)	37 (32.7)	99 (40.6)		
4	4 (6.2)	12 (10.6)	15 (6.1)		
5	0 (0.0)	1 (0.9)	2 (0.8)		
6	0 (0.0)	0 (0.0)	1 (0.4)		

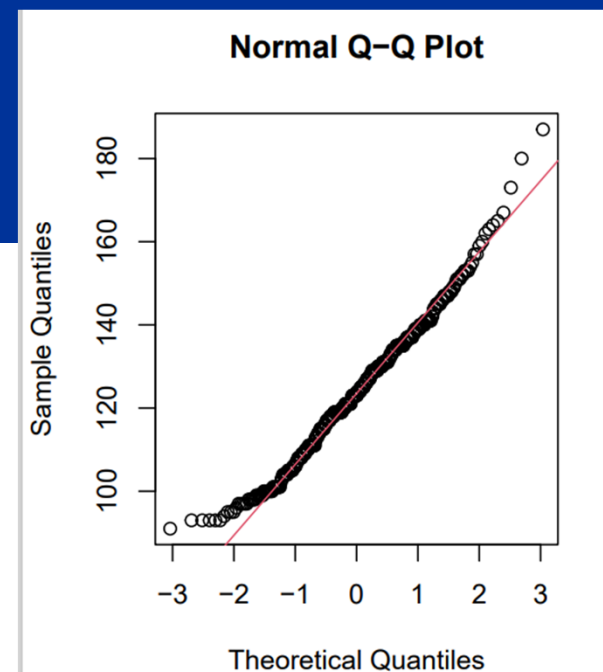


QQプロット

qqnorm(x)

直線を赤 (col=2) で描く

qqline(x, col=2)



One-sample Kolmogorov-Smirnov test

data: scale(x)

D = 0.037474, p-value = 0.5955

alternative hypothesis: two-sided



ks.test(x, "pnorm",
mean=mean(x), sd=sd(x))

Shapiro-Wilk normality test

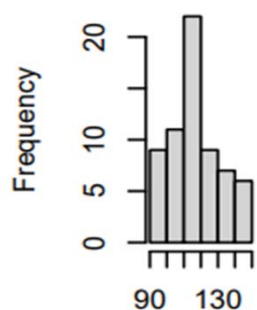
data: x

W = 0.98514, p-value = 0.0002601

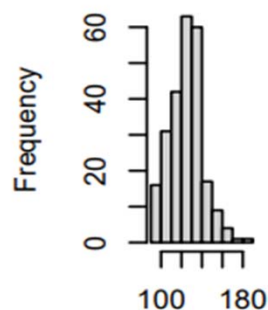


shapiro.test(x)

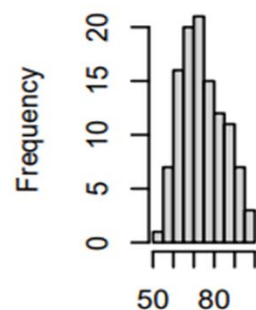
Histogram of $X[[i]]$ Histogram of $X[[i]]$ Histogram of $X[[i]]$



$X[[i]]$

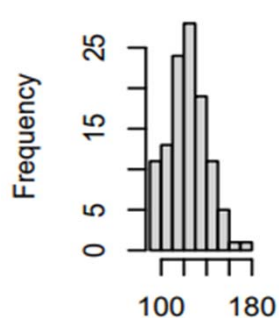


$X[[i]]$

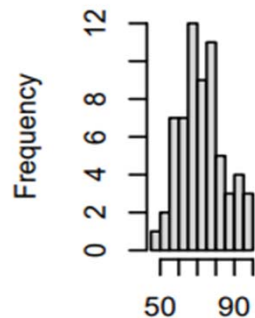


$X[[i]]$

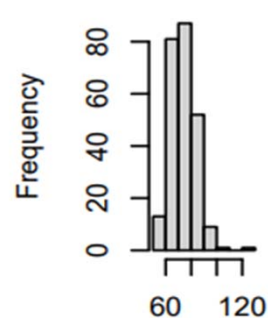
Histogram of $X[[i]]$ Histogram of $X[[i]]$ Histogram of $X[[i]]$



$X[[i]]$



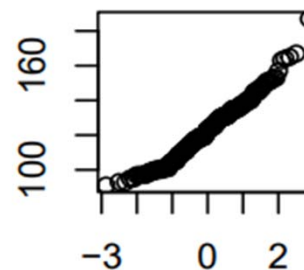
$X[[i]]$



$X[[i]]$

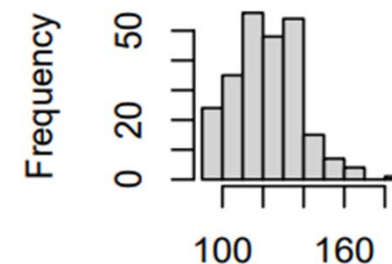
Sample Quantiles

Normal Q-Q Plot



Theoretical Quantiles

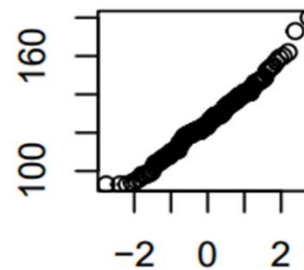
Histogram of $X[[i]]$



$X[[i]]$

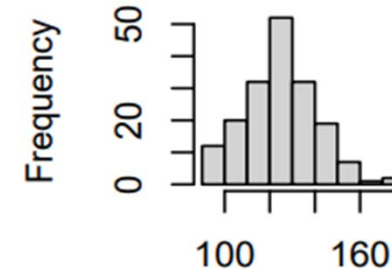
Sample Quantiles

Normal Q-Q Plot



Theoretical Quantiles

Histogram of $X[[i]]$



$X[[i]]$

ggplot2

- `ggplot(data, aes(x= , y=)) +
 geom_ XXXXXXXXXX ()`

Scatter plot: `geom_point()`

Line Chart: `geom_line()`

Histogram: `geom_histogram()`

Box-and-whisker: `geom_boxplot()`

Bar Chart: `geom_bar()`

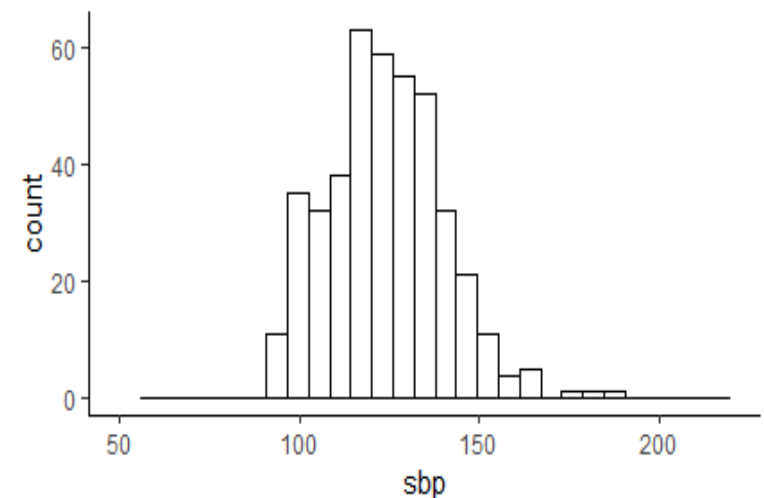
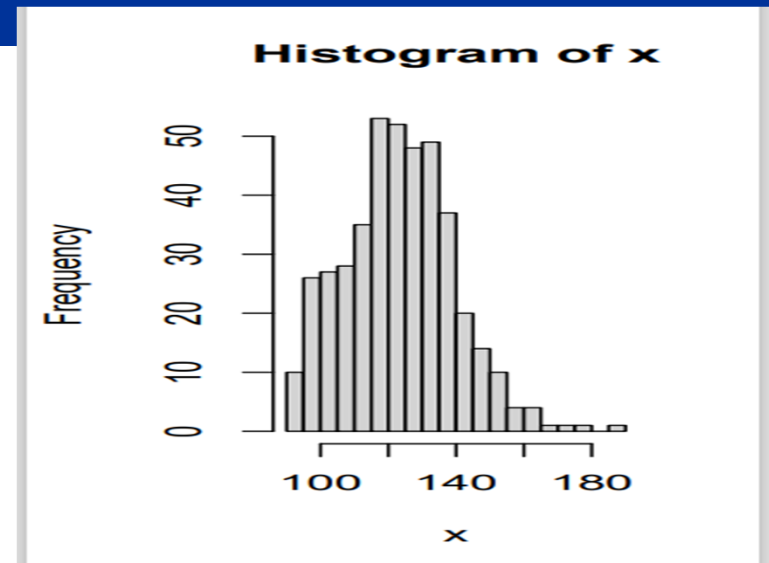
```
# histogram 1
```

```
hist(x, breaks="FD")
```

```
# histogram 2
```

```
g_hist_2 <- ggplot(data= health2, aes(x = sbp))+  
  geom_histogram (fill = "white", color = "black") +  
  theme_classic()+  
  scale_x_continuous(limits = c(50, 220))
```

```
g_hist_2
```



```
# scatter plot 1
```

```
g_point_2 <- ggplot(data= health2, aes(x = age, y =  
sbp))+
```

```
  geom_point () +
```

```
  theme_classic()
```

```
g_point_2
```

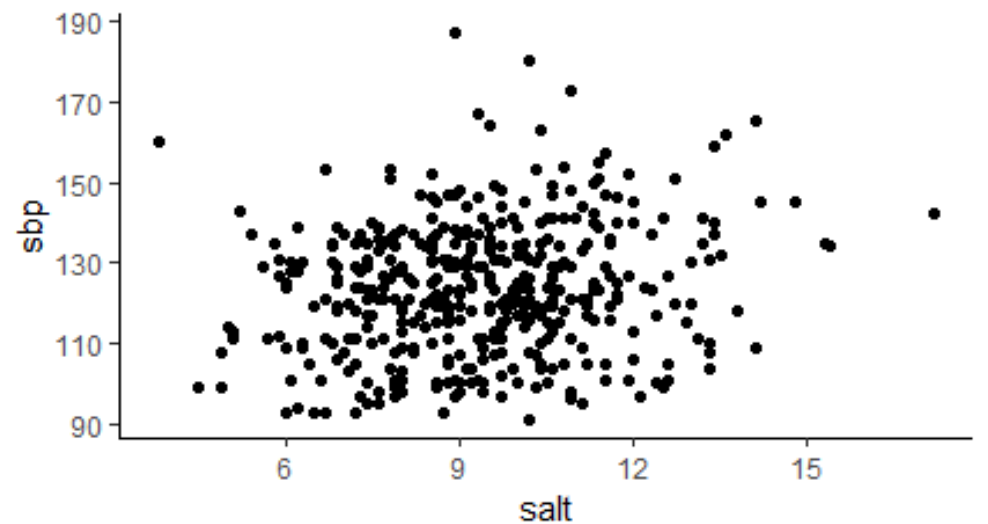
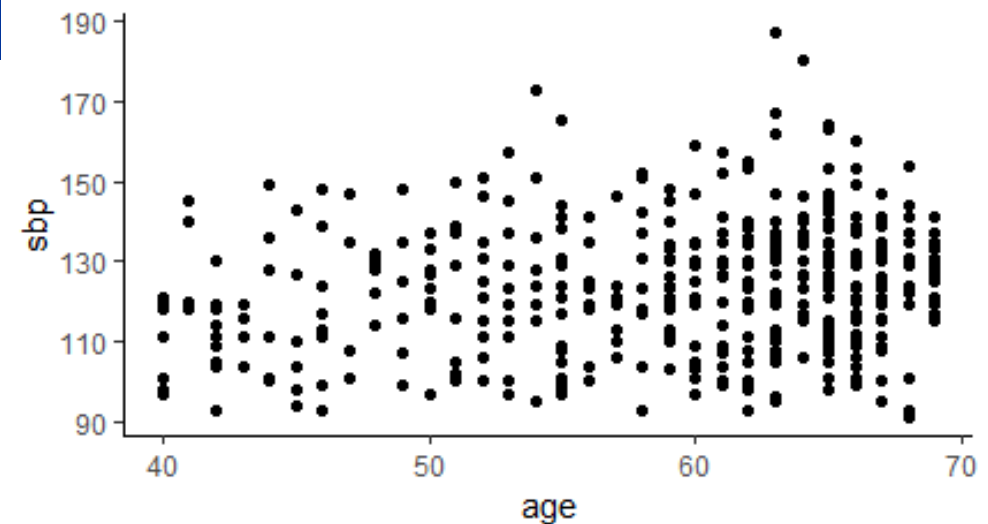
```
# scatter plot 2
```

```
g_point_3 <- ggplot(data= health2, aes(x = salt, y =  
sbp))+
```

```
  geom_point () +
```

```
  theme_classic()
```

```
g_point_3
```



Save Figure

The screenshot displays the RStudio interface. The left pane shows the R script editor with the following code:

```
231 theme_classic()
232
233 g_point_3
234
235 # histogram 1
236 g_hist_2 <- ggplot(data= health2, aes(x = sbp))+
237   geom_histogram (fill = "white", color = "black")+
238   theme_classic()+
239   scale_x_continuous(limits = c(50, 220))
240
241
```

The bottom pane shows the R console with the following output:

```
R 4.1.3 ~/>
+ geom_point () +
+ theme_classic()
>
> g_point_3
> g_hist_2 <- ggplot(data= health2, aes(x = sbp))+
+ geom_histogram (fill = "white", color = "black") +
+ theme_classic()+
+ scale_x_continuous(limits = c(50, 220))
>
```

The right pane shows the Environment tab with the following data objects:

Object	Type	Size
g_hist_2	List of 9	
g_point_2	List of 9	
g_point_3	List of 9	

The bottom right pane shows a scatter plot of 'sbp' (systolic blood pressure) on the y-axis (ranging from 90 to 190) against 'salt' on the x-axis (ranging from 6 to 15). The plot is a histogram with white fill and black borders. A red arrow points from the code editor to the plot. A red box highlights the 'Export' menu, which includes the following options:

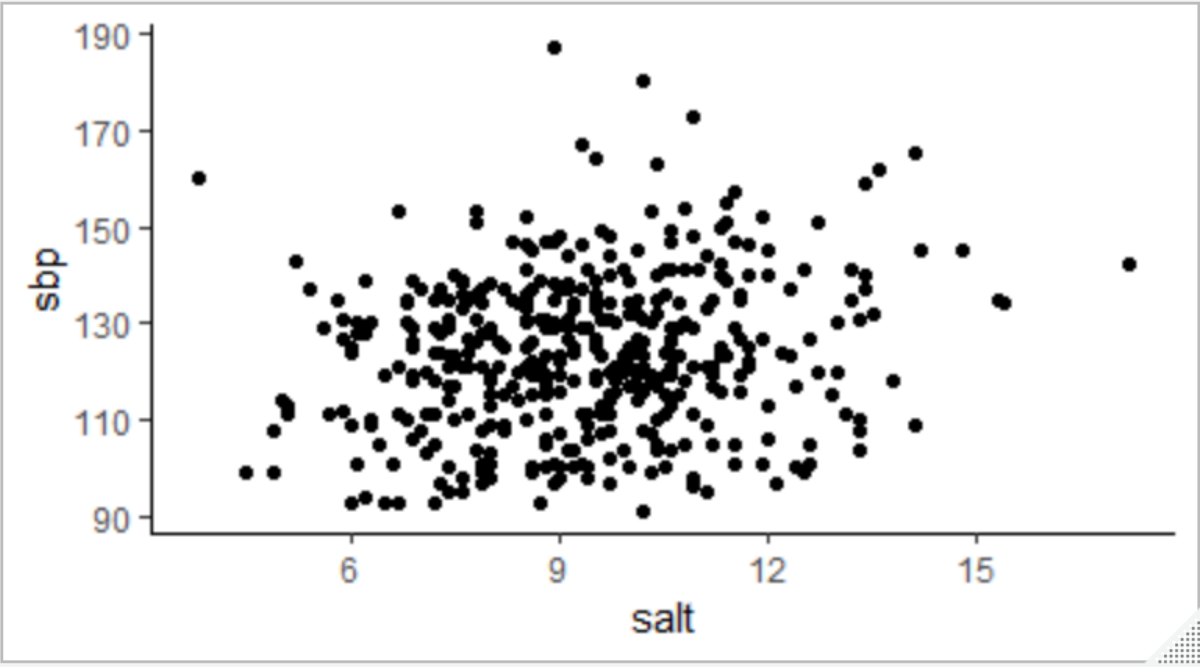
- Save as Image...
- Save as PDF...
- Copy to Clipboard...

Save Plot as Image

Image format: PNG Width: 430 Height: 236

Directory... D:/ ☐ Maintain aspect ratio

File name: Rplot14 Update Preview



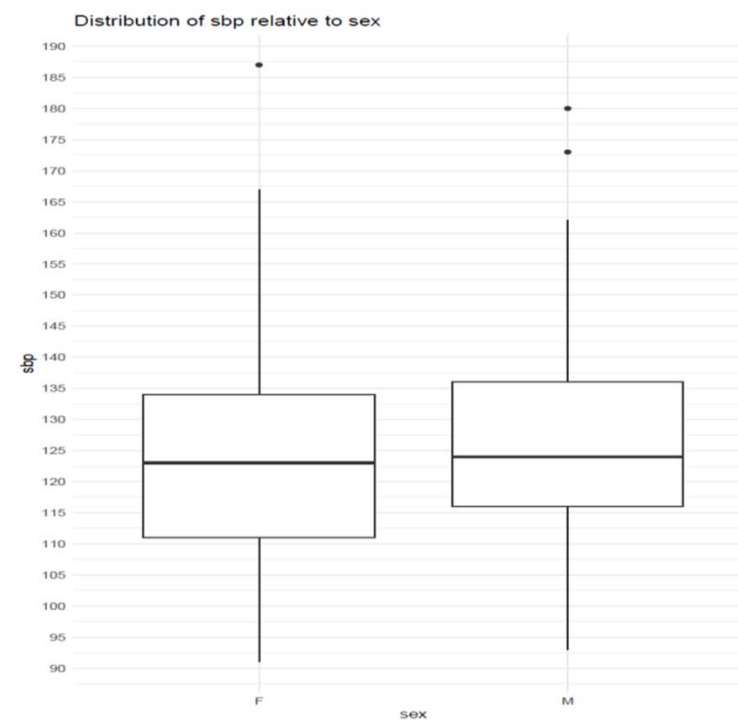
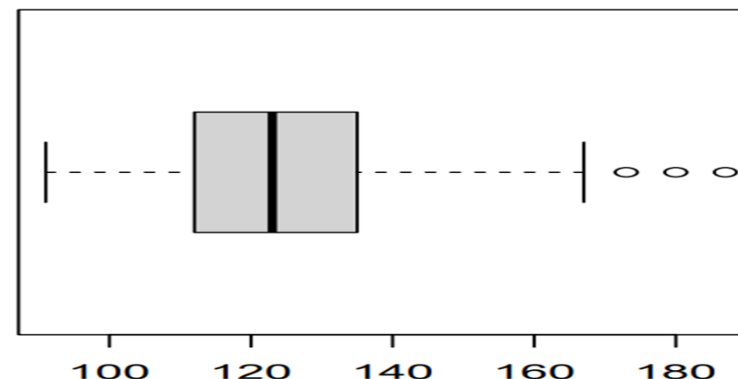
☐ View plot after saving Save Cancel

The scatter plot displays the relationship between salt intake (x-axis) and systolic blood pressure (sbp, y-axis). The x-axis, labeled 'salt', ranges from approximately 5 to 18, with major tick marks at 6, 9, 12, and 15. The y-axis, labeled 'sbp', ranges from 90 to 190, with major tick marks at 90, 110, 130, 150, 170, and 190. The plot contains numerous black data points, showing a positive correlation between salt intake and blood pressure. The points are most densely clustered between salt values of 6 and 12 and sbp values of 110 and 150.

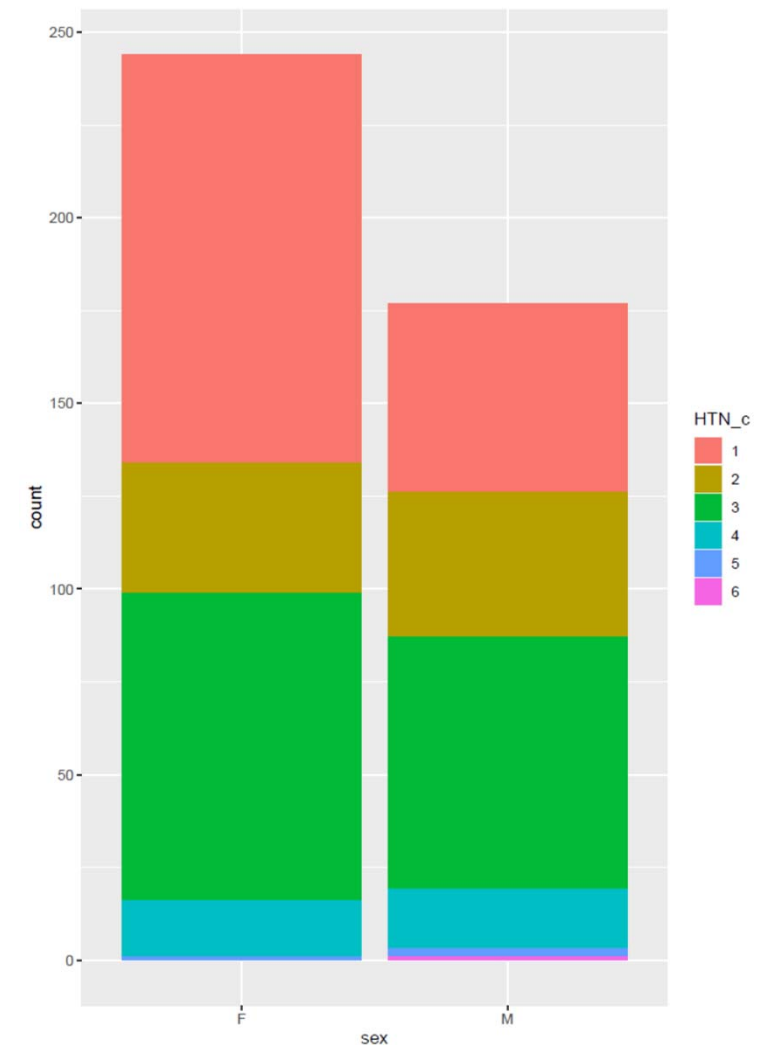
Box plot

```
#boxplot 1  
boxplot(x)  
#horizontal=TUREで横向きになる  
boxplot(x, horizontal=TRUE)
```

```
#boxplot 2  
health2 %>%  
  ggplot(aes(sex,sbp)) +  
  geom_boxplot() +  
  labs(title = "Distribution of cty relative to drv",  
        x = "sex",  
        y = "sbp") +  
  theme_minimal() +  
  scale_y_continuous(breaks = seq(70,200,10))
```



```
> health2$HTN_c<-  
as.character(health2$HTN_c)  
> health2 %>%  
+ ggplot(aes(sex)) +  
+ geom_bar(aes(fill = HTN_c))  
>
```

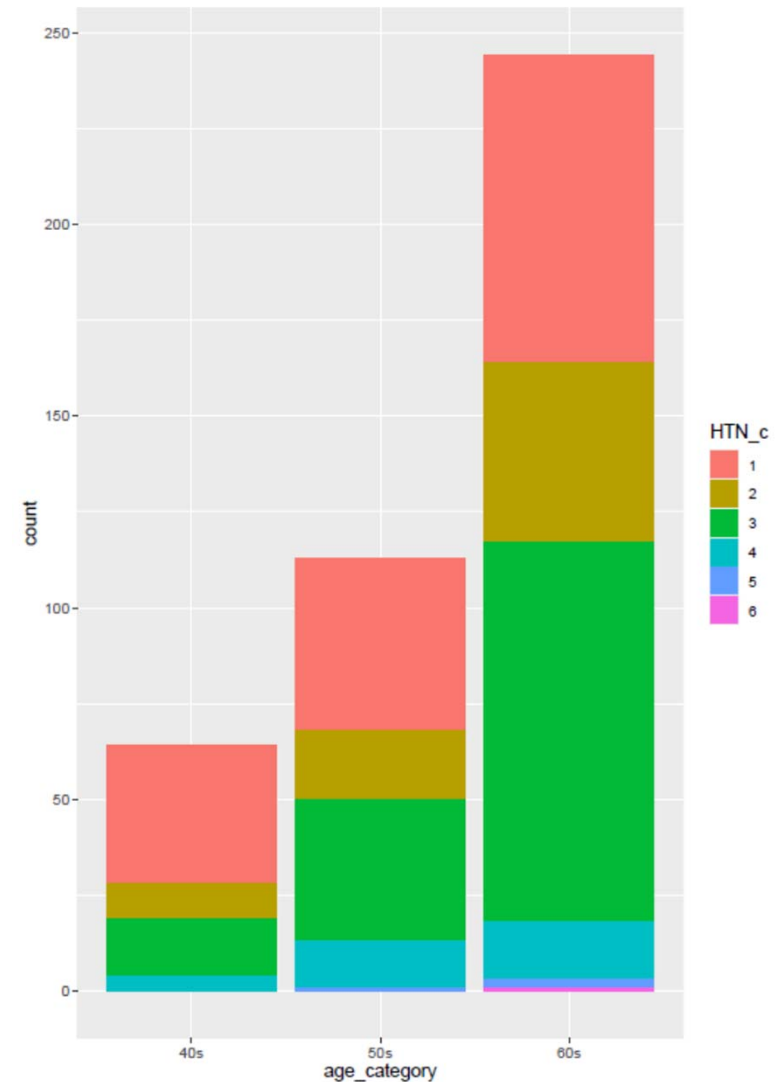



```

> health2x <-health2 %>%
+ mutate(age_category = case_when(age < 30 ~
"20s",
+ age >= 30 & age < 40 ~ "30s",
+ age >= 40 & age < 50 ~ "40s",
+ age >= 50 & age < 60 ~ "50s",
+ age >= 60 & age < 70 ~ "60s",
+ age >= 70 & age < 80 ~ "70s",
+ age >= 80 ~ "80s"
+ )
+ )
> head(health2x)

> health2x %>%
+ ggplot(aes(age_category)) +
+ geom_bar(aes(fill = HTN_c))

```



相関

- 「health2.csv」を使用
- 血圧データを使用し、相関のコード例を示してある

相関

#相関行列表 (correlation matrix) を出す
場合

```
attach(health2)  
cor(cbind(sbp, dbp))
```

#相関係数以外 (p値など) も出す場合
cor.test(sbp, dbp)

#スピアマンの順位相関係数
cor(sbp, dbp, method="spearman")

#ケンドールの順位相関係数 (データが少
なくて, 同率順位が多いとき)
cor(sbp, dbp, method="kendall")

```
> health2 %>% select (age, sbp, dbp,  
salt) %>%  
+ cor %>%  
+ pander
```

```
-----  
&nbsp; age    sbp    dbp    salt  
-----  
**age**   1    0.1686 0.0174 0.0891  
  
**sbp**  0.1686   1    0.7189 0.1776  
  
**dbp**  0.0174 0.7189   1    0.1537  
  
**salt** 0.0891 0.1776 0.1537   1  
-----
```



相関

```
quant<-cbind  
(health2$age, health2$sbp,  
health2$dbp,  
health2$salt)  
quant2<-cbind (age, sbp,  
dbp, salt)  
cor(quant2)  
pairs(quant2)
```

